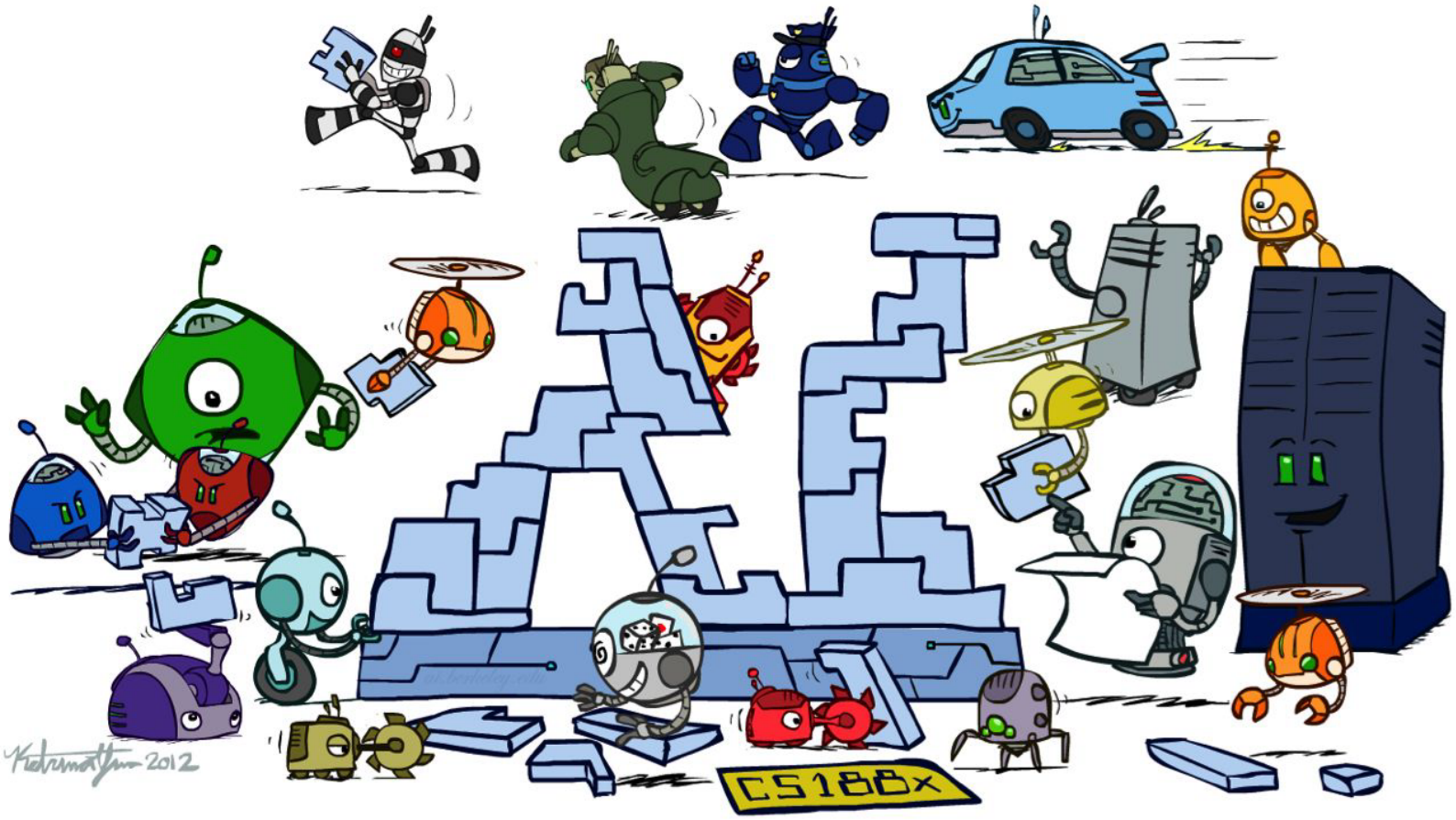
Conclusion



[These slides were created by Cam Allen, Michael Cohen, Dan Klein and Pieter Abbeel for CS188 Intro to AI at UC Berkeley.

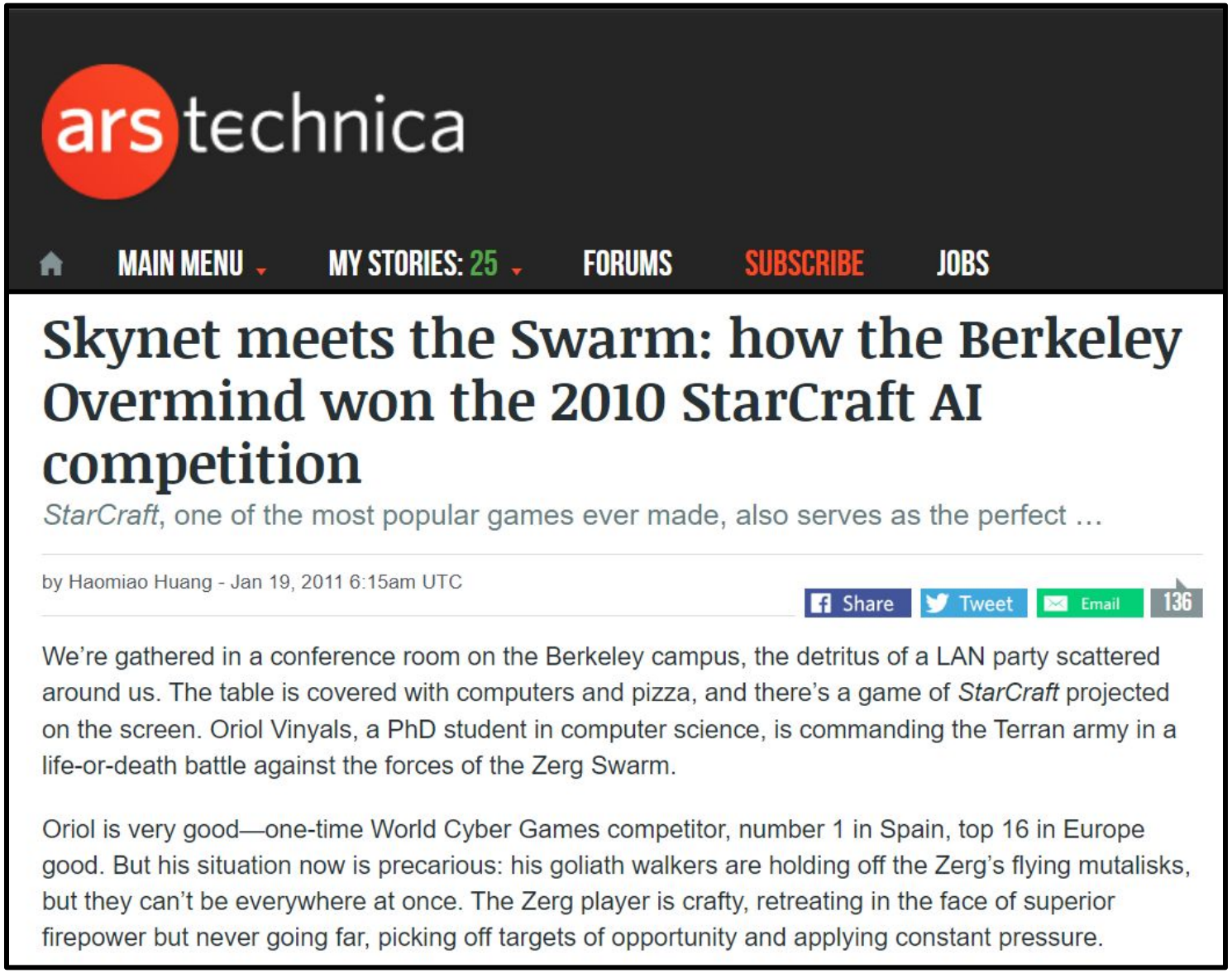
All CS188 materials are available at http://ai.berkeley.edu.]

Core AI Ideas

- We've seen some key ideas about how computation can give rise to aspects of intelligent behavior
  - Planning: Search, CSPs, Games (predicting outcomes in simulation)
  - Uncertainty: MDPs, Bayes Nets, Utilities (decision making with uncertainty)
  - Learning: ML and RL: (improving from experience)

- Today: How might these fit together?

The Berkeley Overmind Project (2011)



ars technica

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Skynet meets the Swarm: how the Berkeley Overmind won the 2010 StarCraft AI competition

StarCraft, one of the most popular games ever made, also serves as the perfect …

by Haomiao Huang - Jan 19, 2011 6:15am UTC

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We're gathered in a conference room on the Berkeley campus, the detritus of a LAN party scattered around us. The table is covered with computers and pizza, and there's a game of *StarCraft* projected on the screen. Oriol Vinyals, a PhD student in computer science, is commanding the Terran army in a life-or-death battle against the forces of the Zerg Swarm.

Oriol is very good—one-time World Cyber Games competitor, number 1 in Spain, top 16 in Europe good. But his situation now is precarious: his goliath walkers are holding off the Zerg's flying mutalisks, but they can't be everywhere at once. The Zerg player is crafty, retreating in the face of superior firepower but never going far, picking off targets of opportunity and applying constant pressure.

STARCRAFT



StarCraft is a trademark of Blizzard Entertainment, Inc.

STARCRAFT

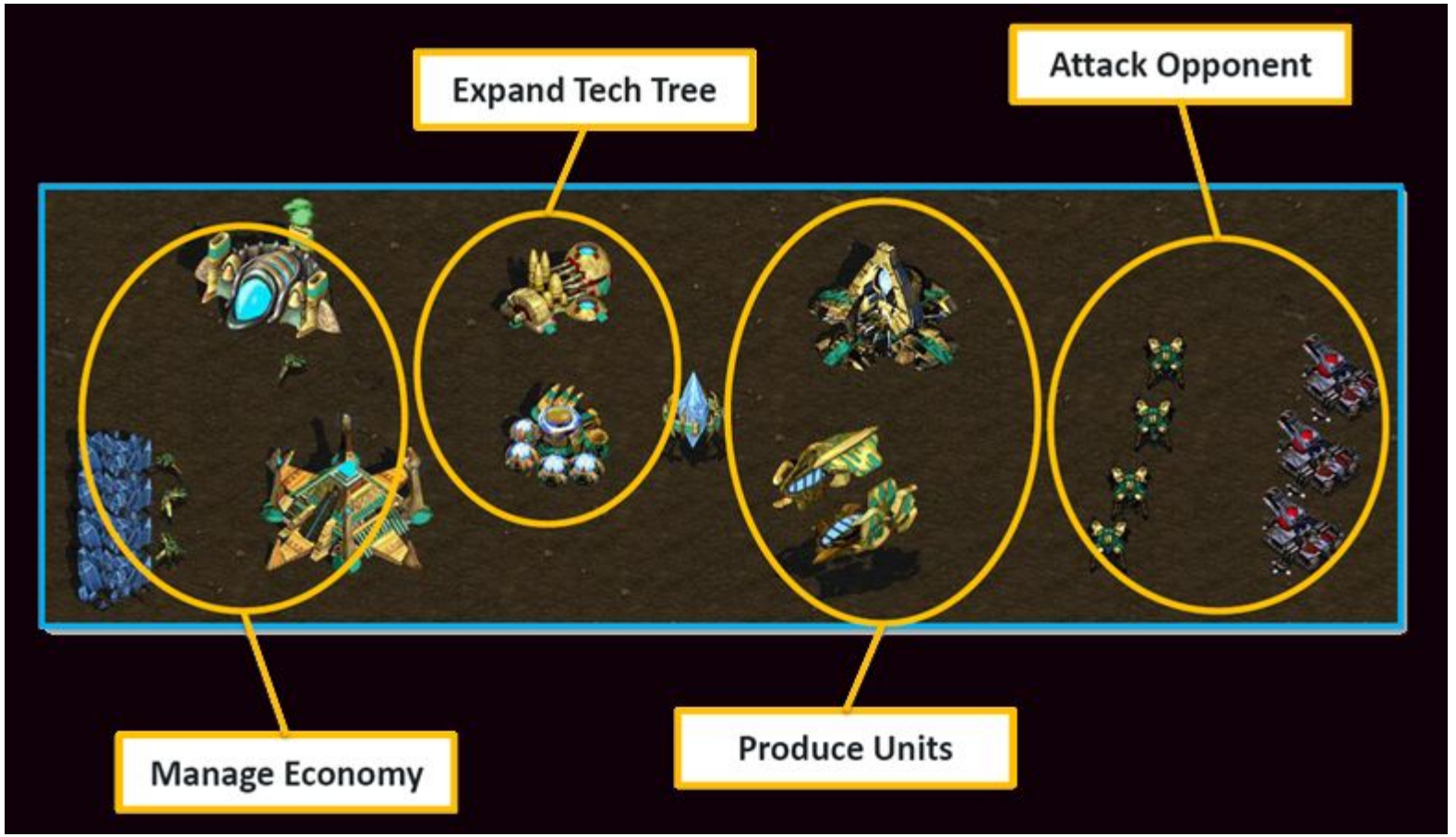


Image from Ben Webber

Why is Starcraft Hard for AI?

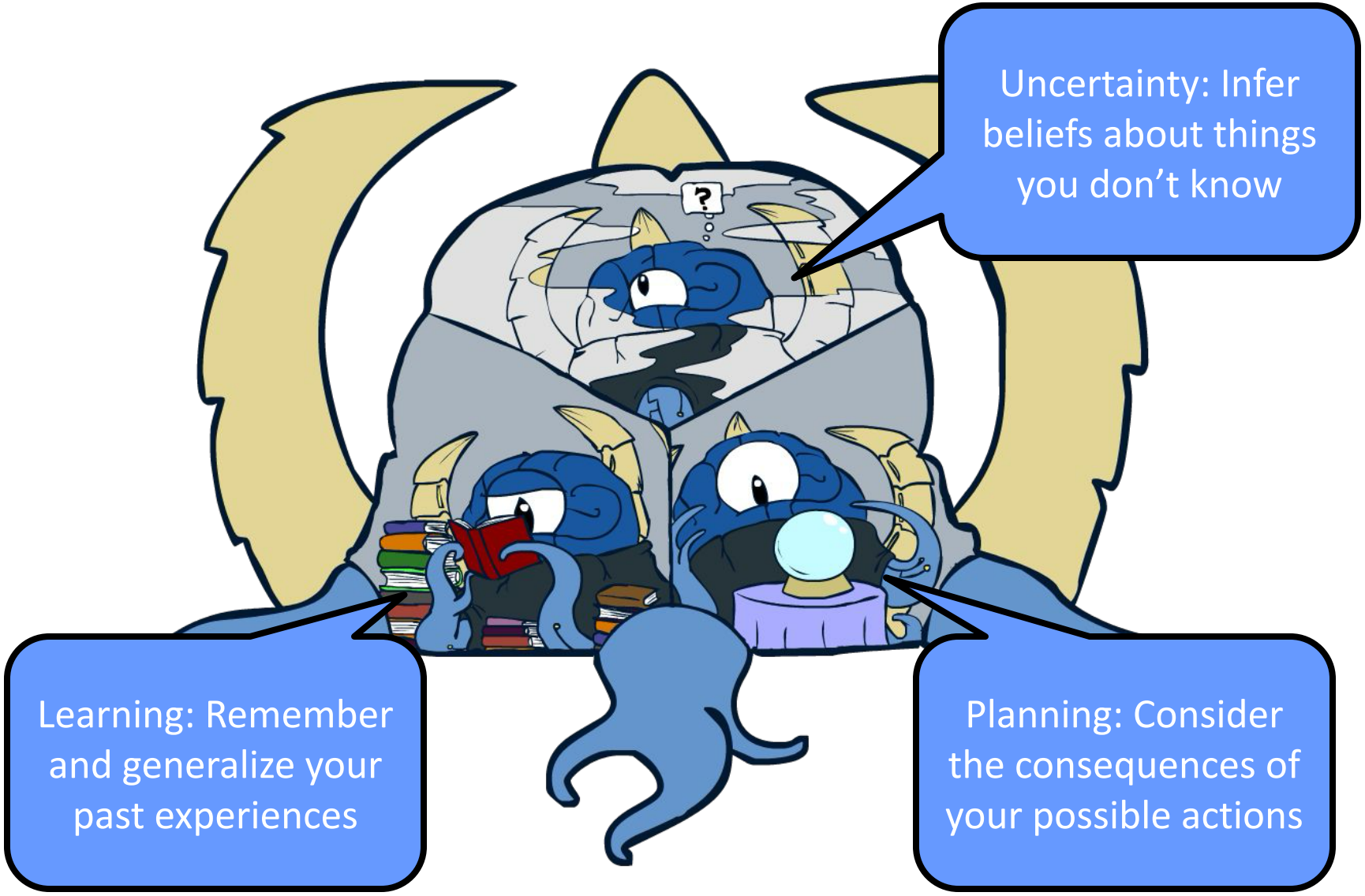
- **The game of Starcraft is:**
  - Adversarial
  - Long Horizon
  - Partially Observable
  - Realtime
  - Huge branching factor
  - Concurrent
  - Resource-rich
  - …

- **No single CS188 algorithm (e.g. minimax) will solve it off-the-shelf**





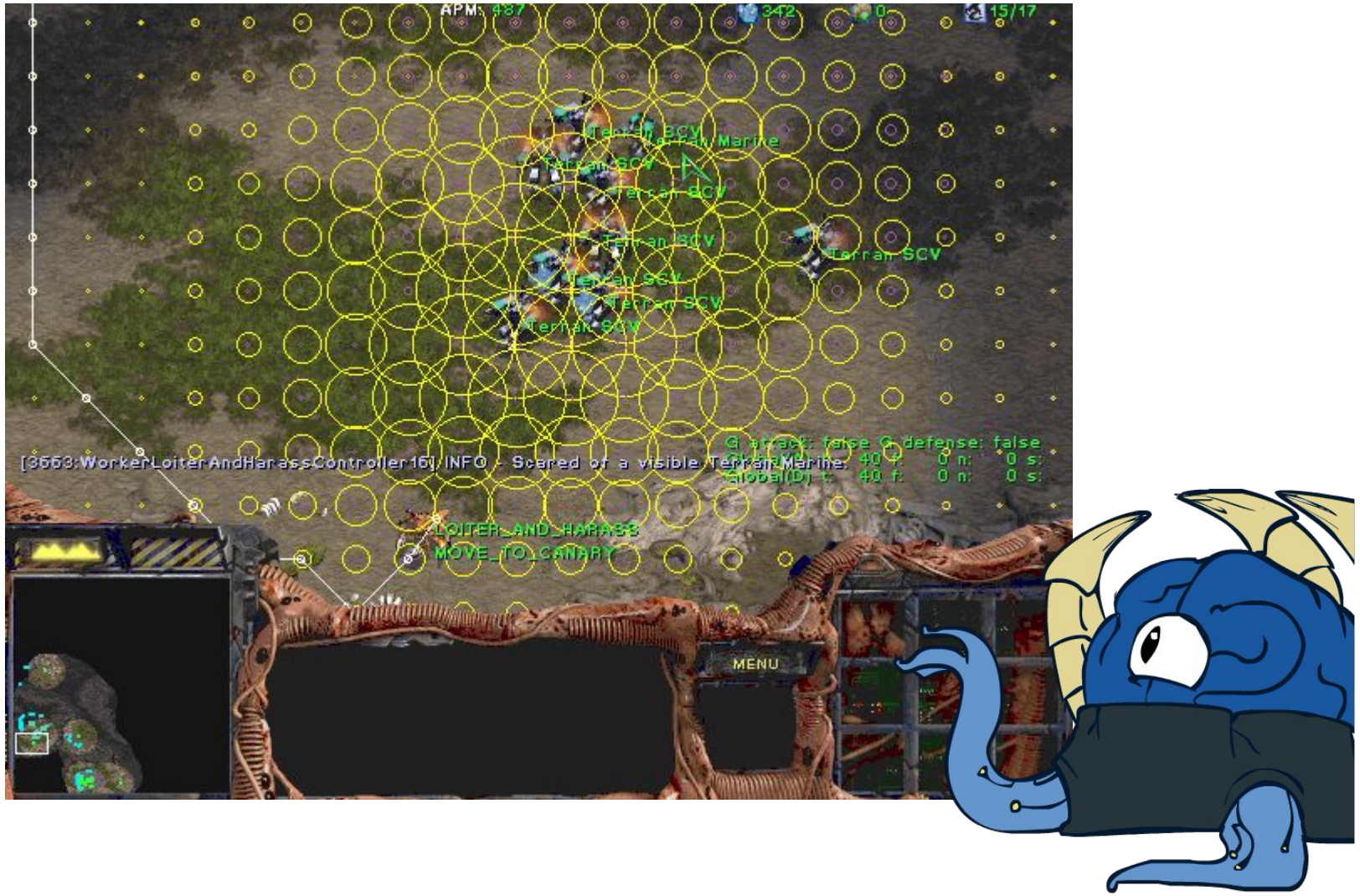
Three Key Ideas in AI



Planning



Search

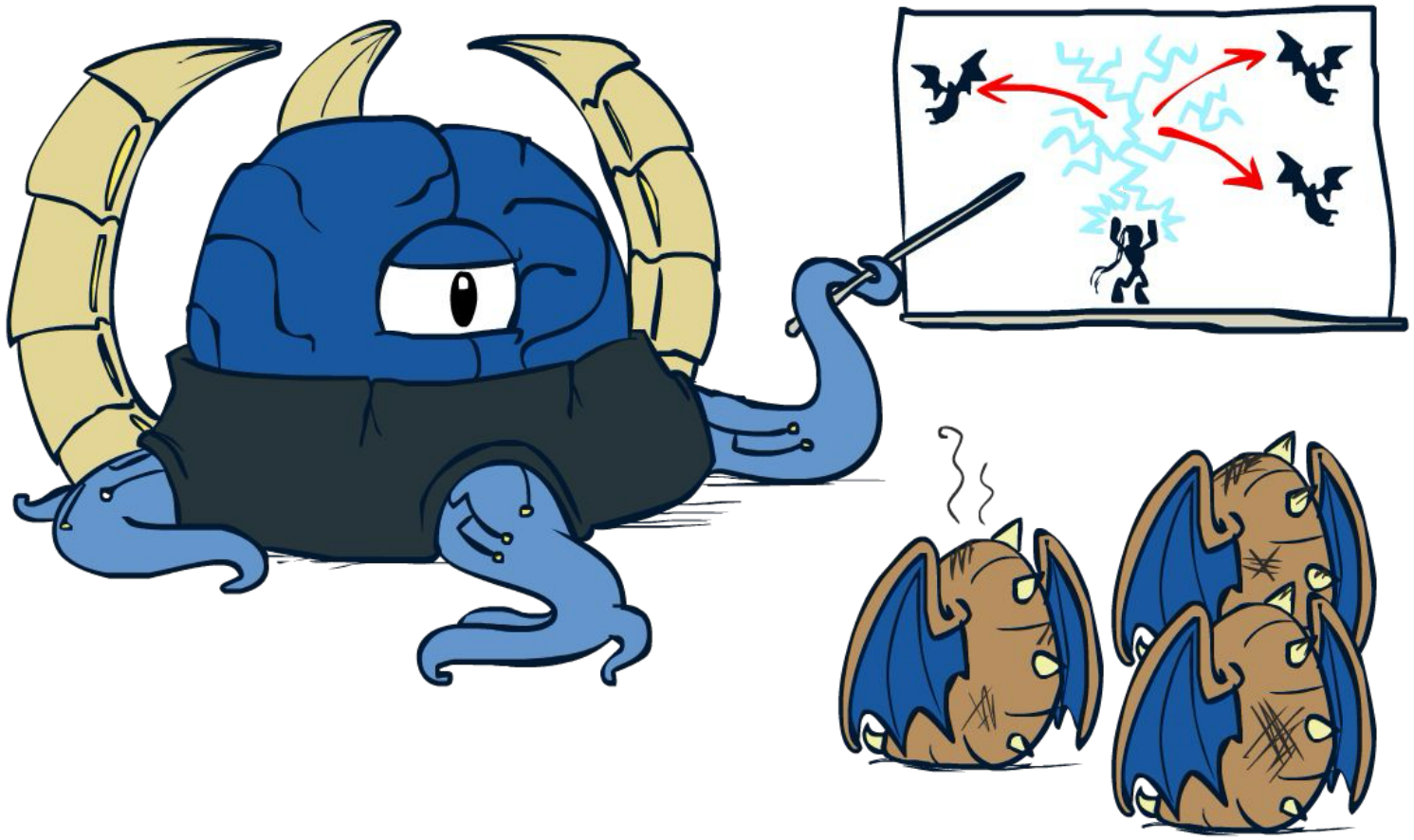

APM: 487
342
0
15/17
Terran SCV
Terran Marine
Terran SCV
Terran SCV
Terran SCV
Terran SCV
Terran SCV
Terran SCV
Terran SCV
[3663:WorkerLoiterAndHarassController16]INFO - Scared of a visible Terran Marine
G attack: false G defense: false
Global(A) t: 40 f: 0 n: 0 s:
Global(D) t: 40 f: 0 n: 0 s:
LOITER_AND_HARASS
MOVE_TO_CANARY
MENU



Game Trees


APM: 6744
195
224
66/76
G attack: false G defense: true
Global(A) t: 1171 f: 0 n: 300 s:
Global(D) t: 151 f: 897 n: 300 s:
Zerg Lair
MENU
Control Used: 66
Control Provided: 1
Total Control: 76
Control Max: 200
1900/1900

Learning

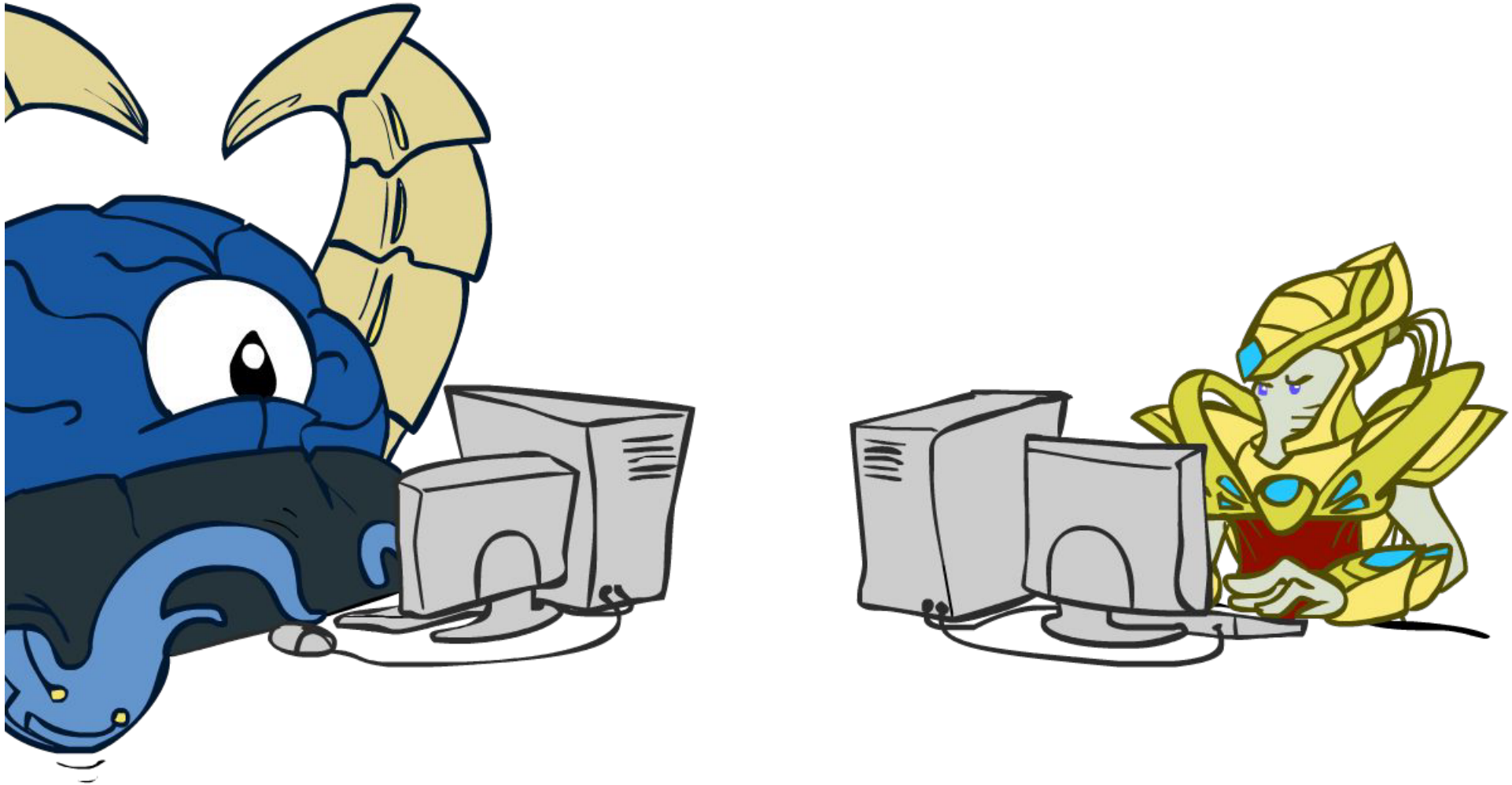


Uncertainty





AIIDE Starcraft AI Competition

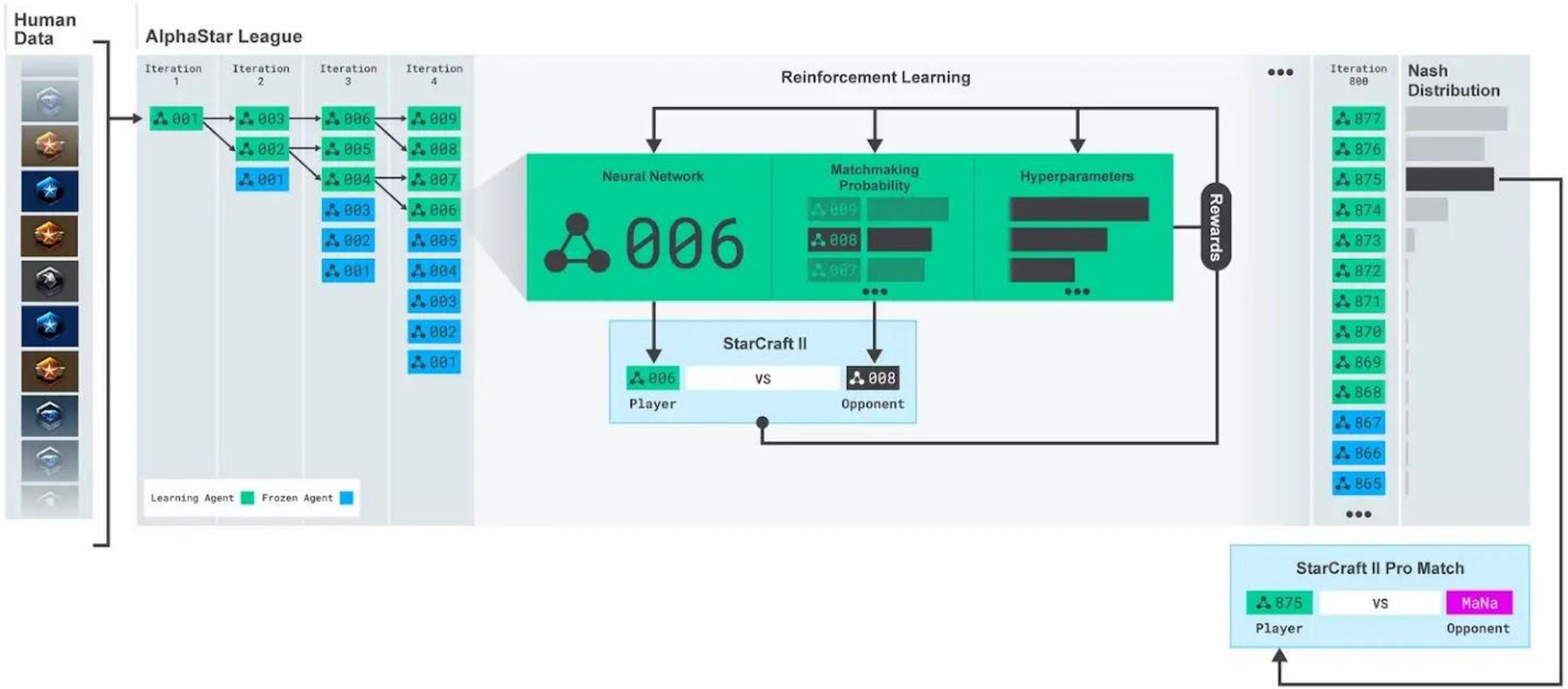


AlphaStar (2019)

- Deepmind (2019) built an agent for Starcraft II
- Learned to play based on neural nets, rollouts, and population learning
- Outplayed the best human players!



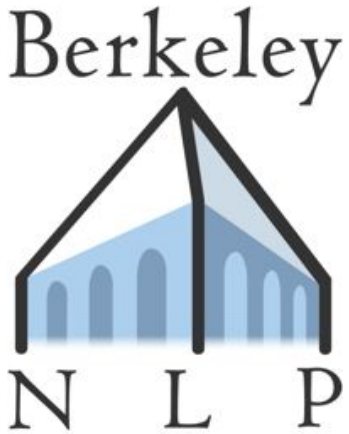
AlphaStar: Population Self-Play



AlphaStar (2019)

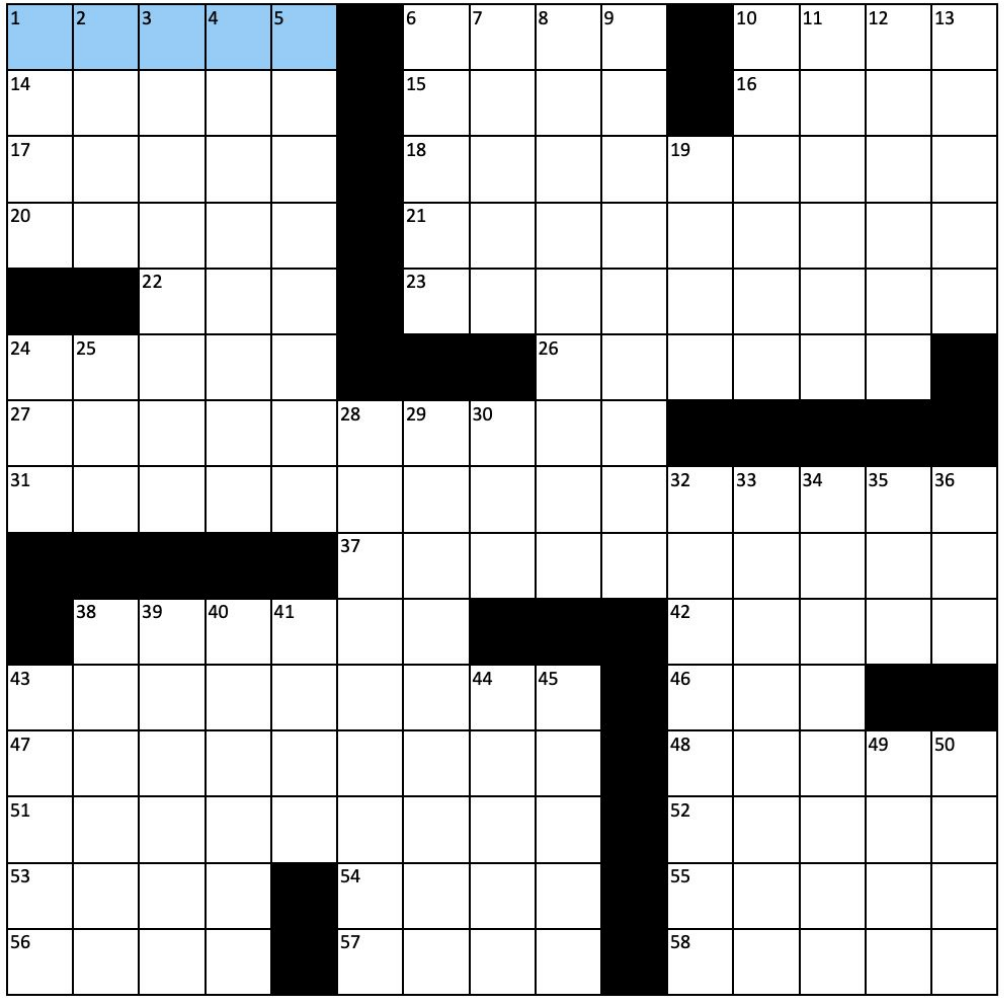


Automated Crossword Solving (2021)



Eric Wallace, Nicholas Tomlin, Albert Xu, Kevin Yang, Eshaan Pathak, Matthew L. Ginsberg, Dan Klein

Example Crossword



Across

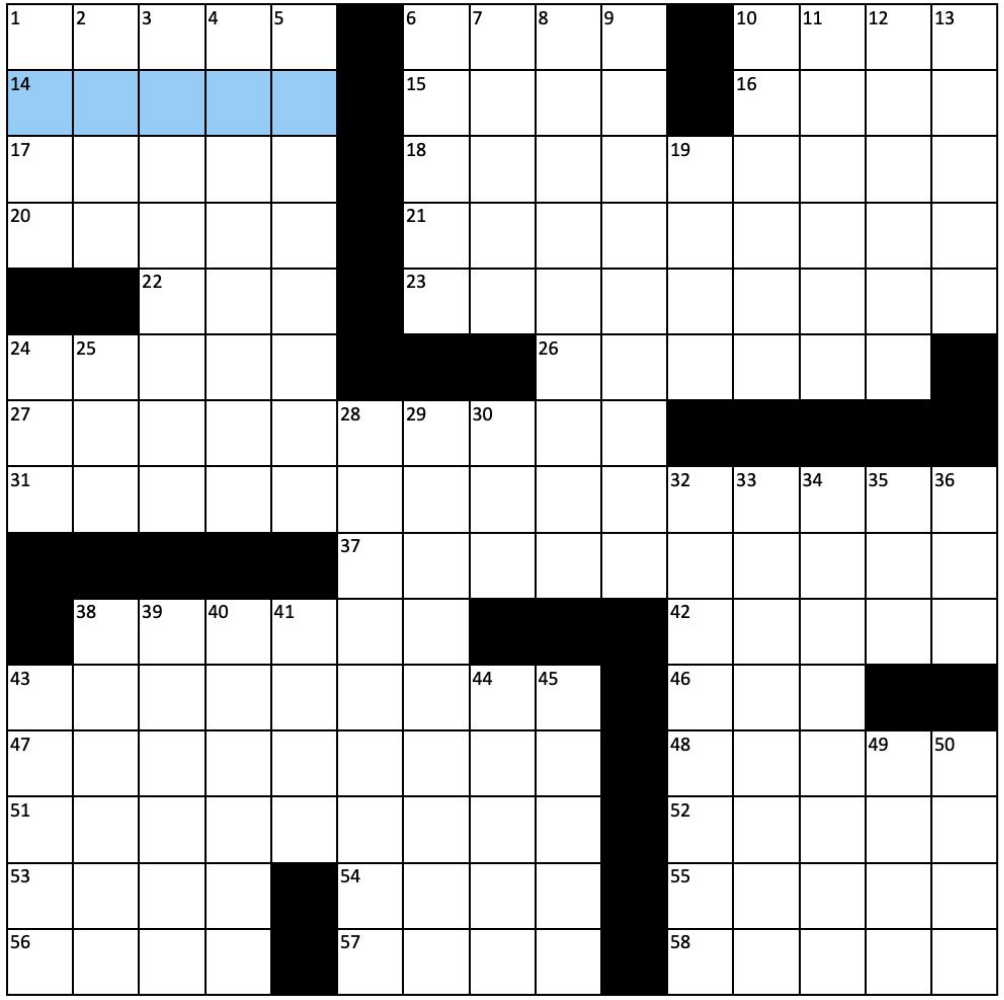
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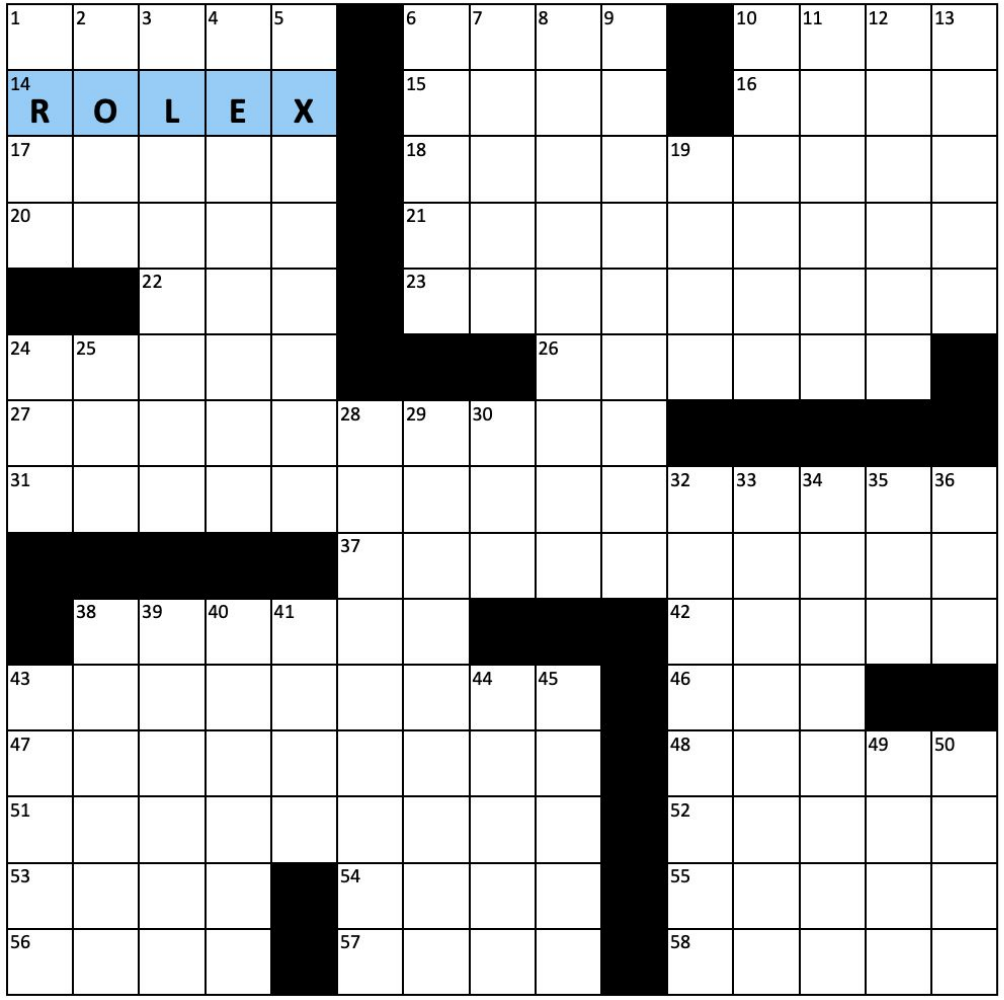
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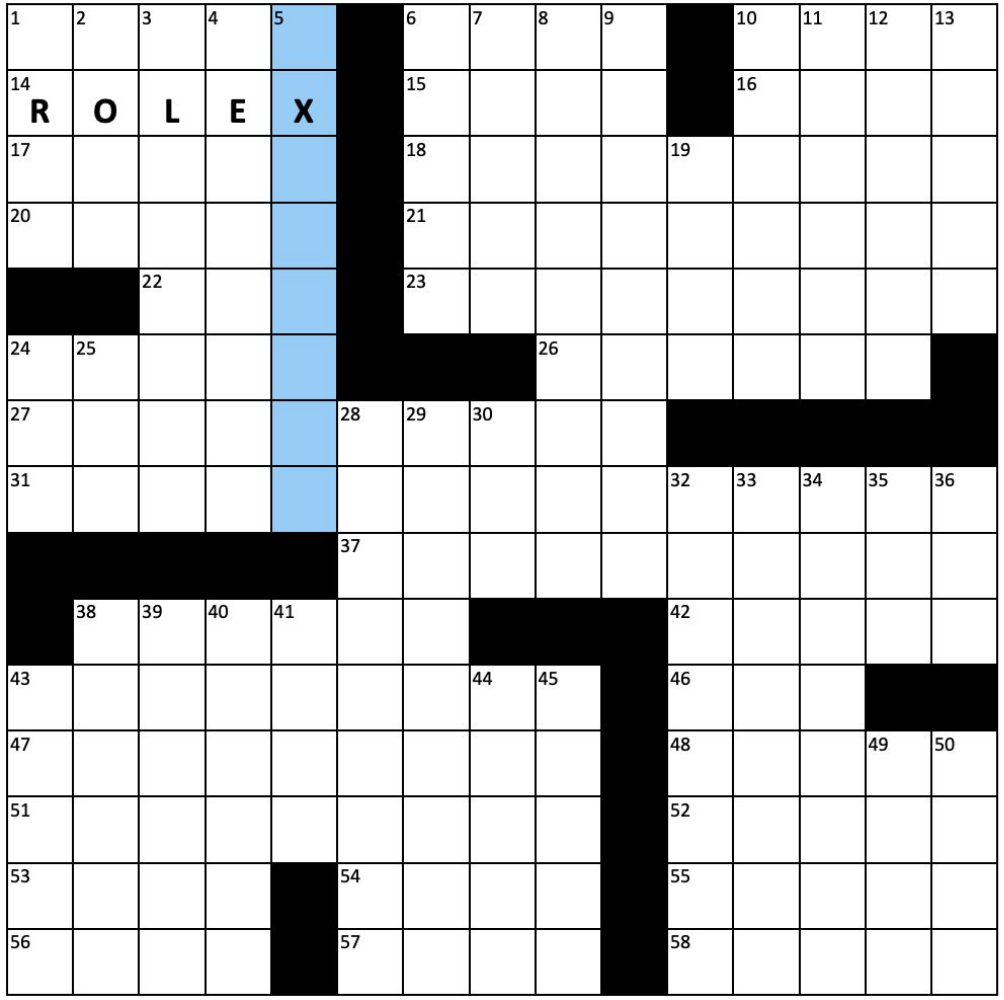
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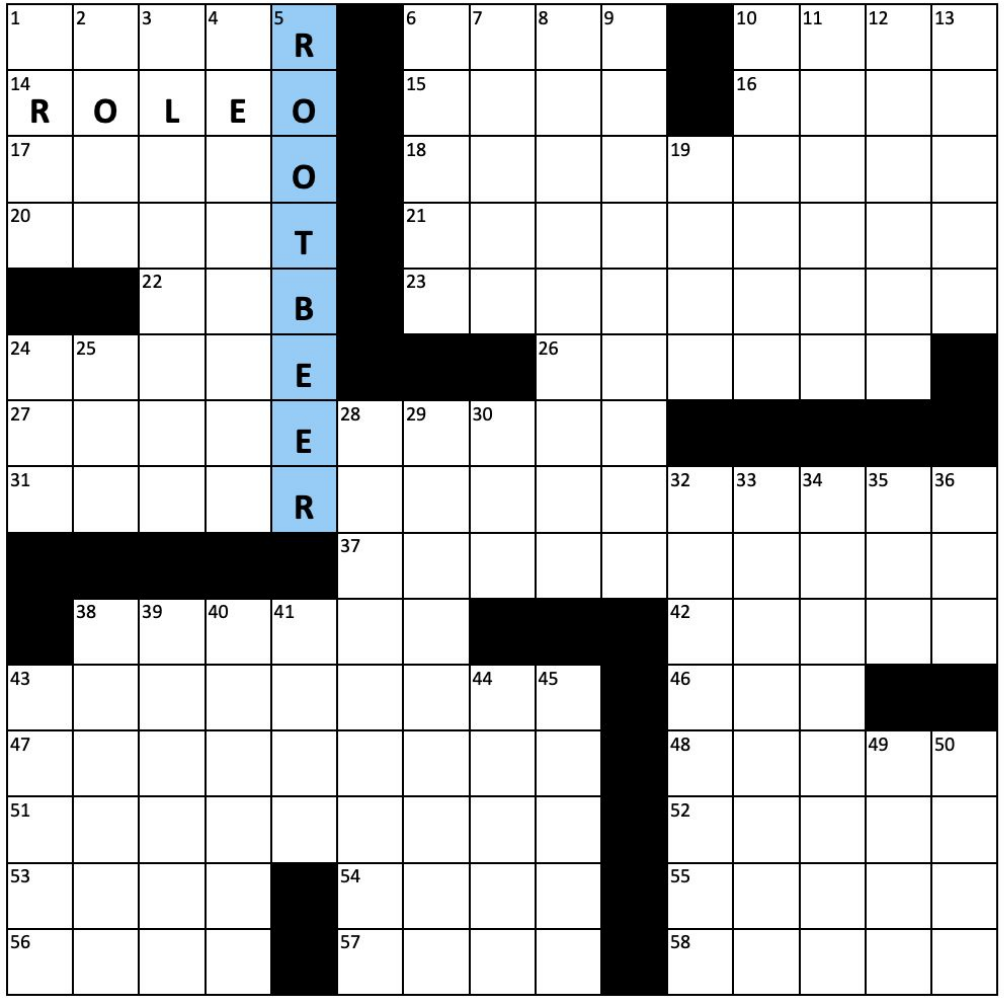
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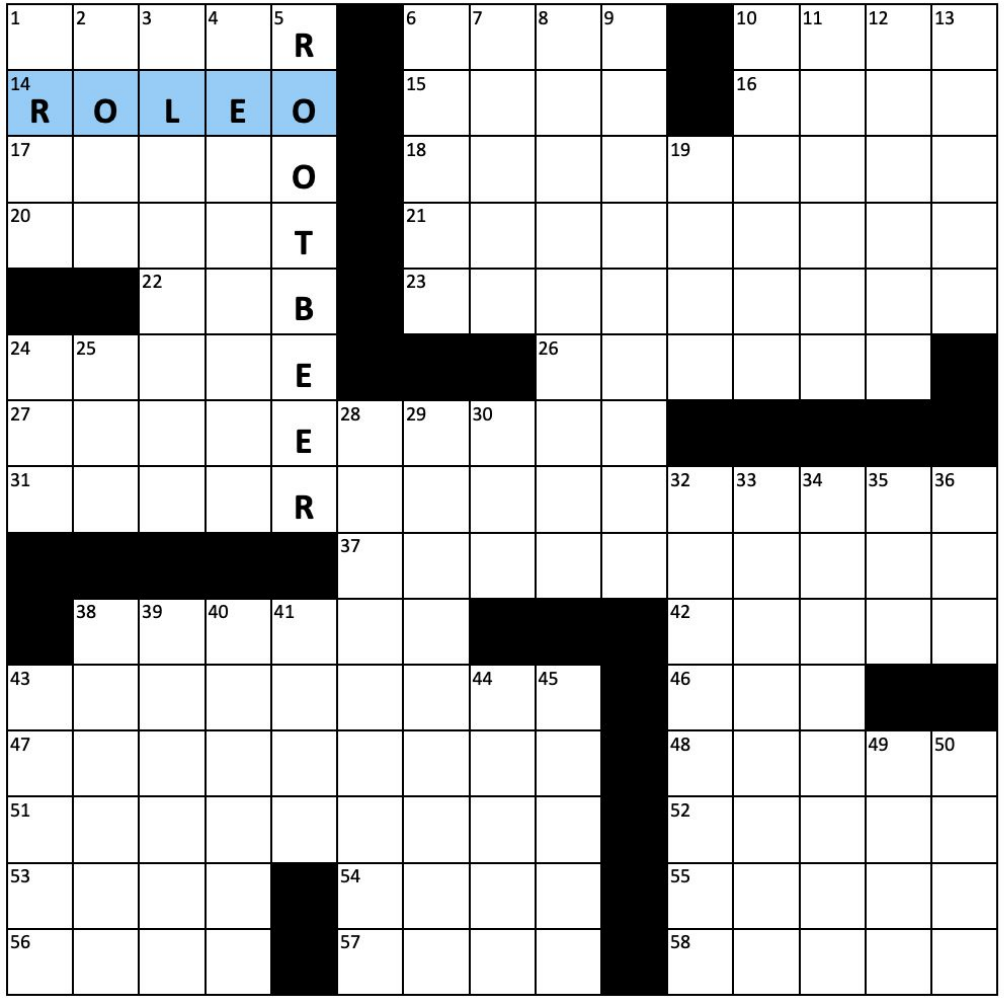
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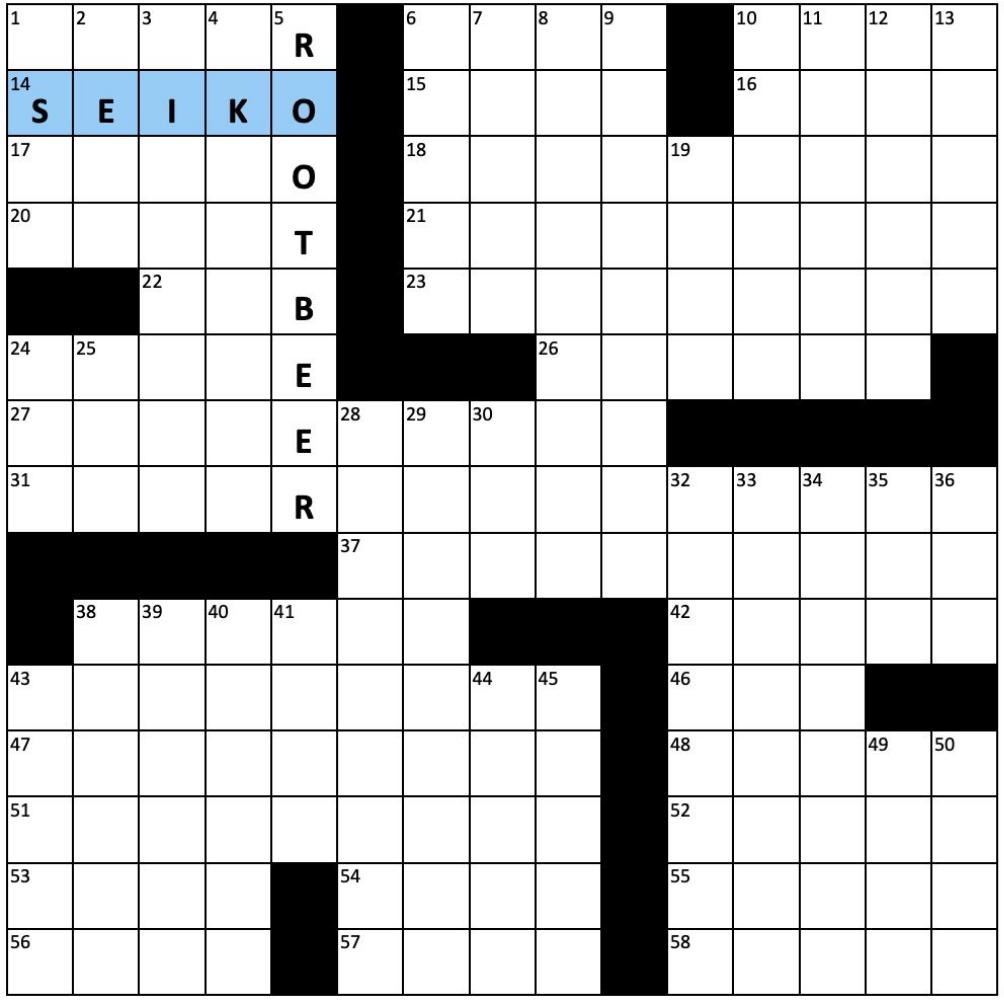
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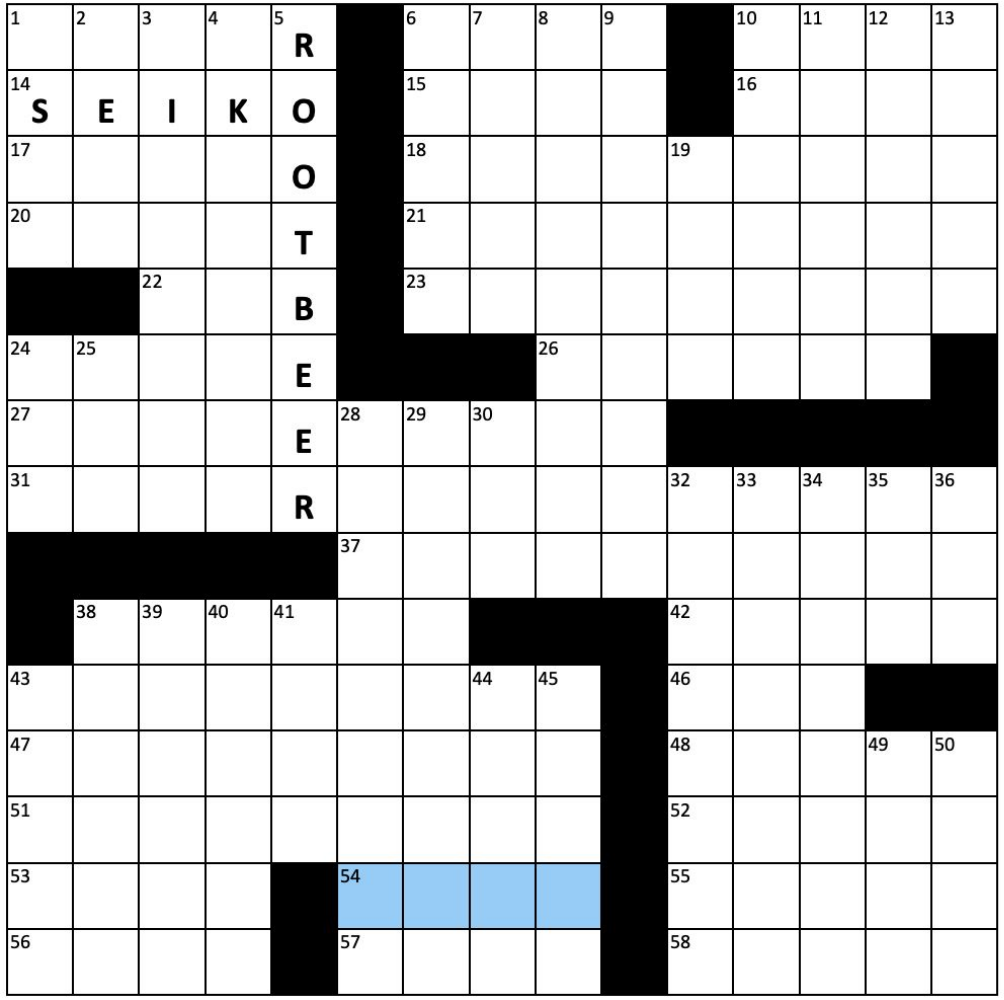
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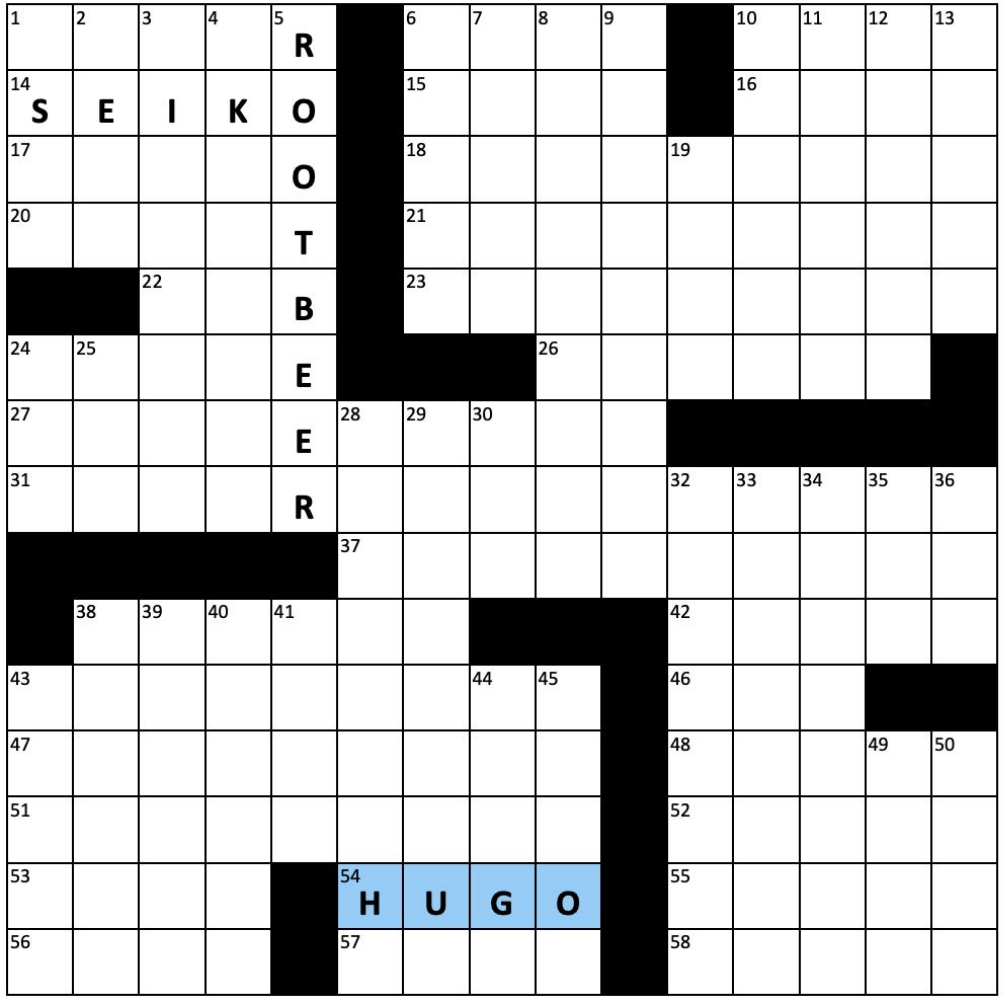
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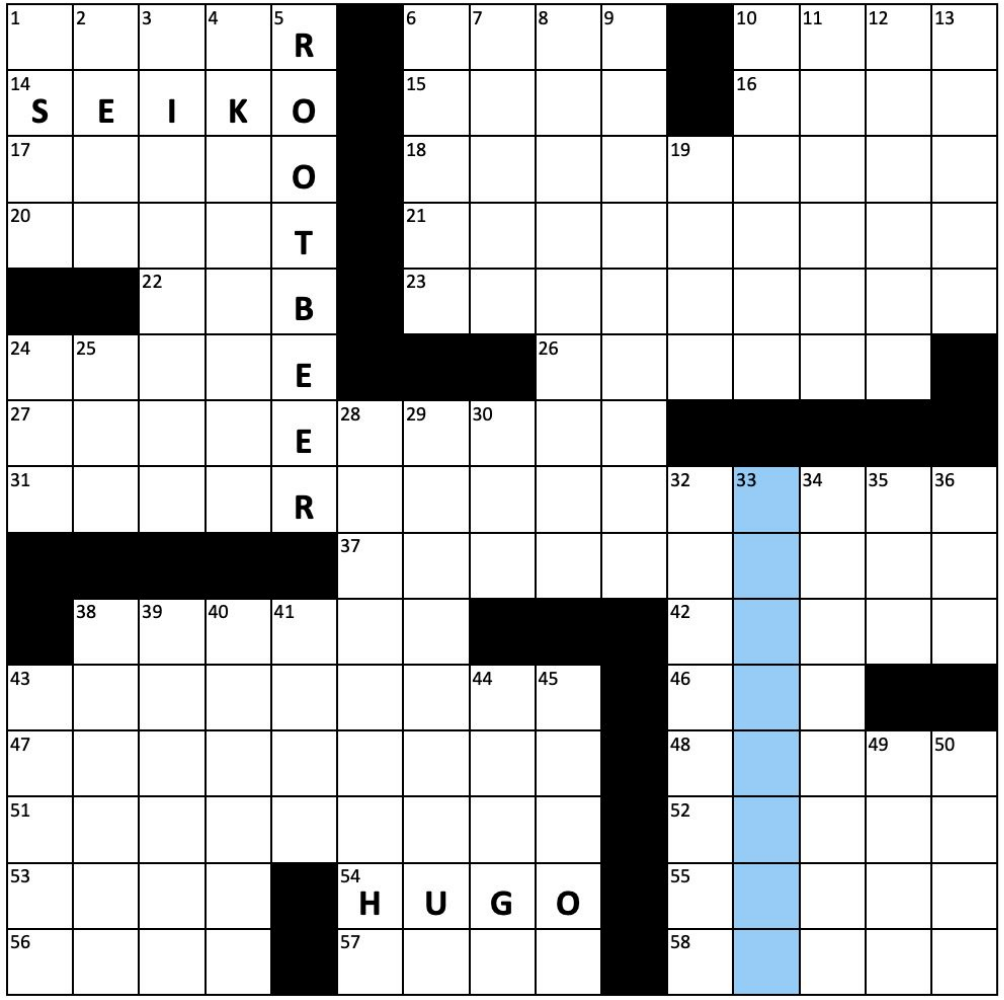
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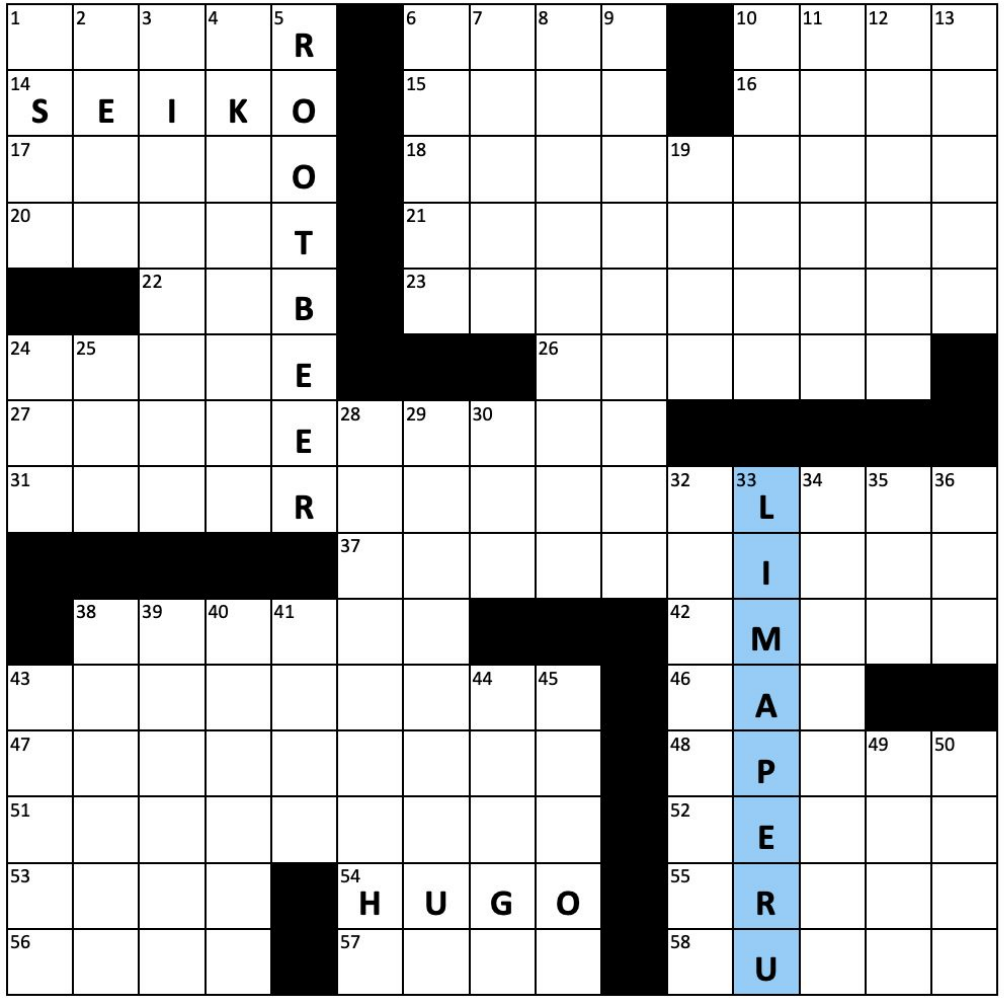
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- **30** It can be found right on a clock
- **32** "100 Years ... 100 Stars" and others, in brief
- **33** The world's largest desert capital, after Cairo, Egypt

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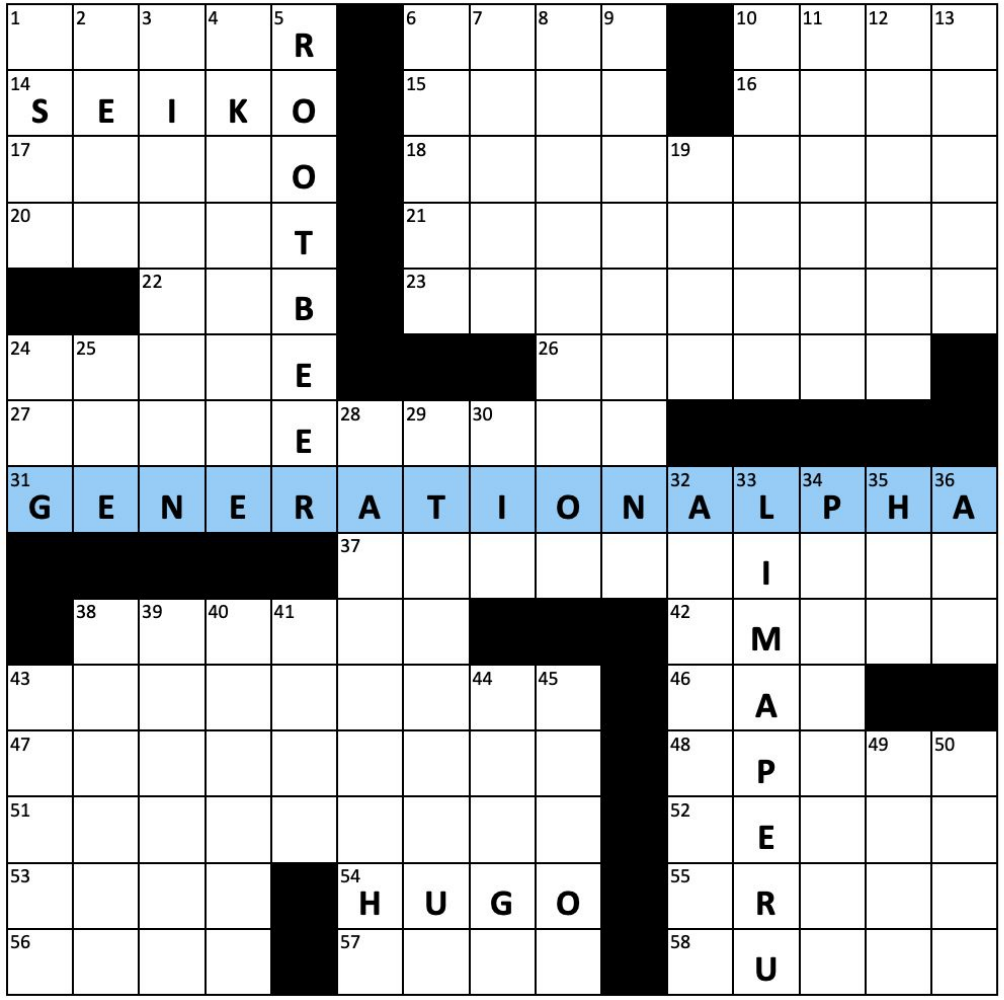
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- **5** Drink similar to sarsaparilla
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Example Crossword



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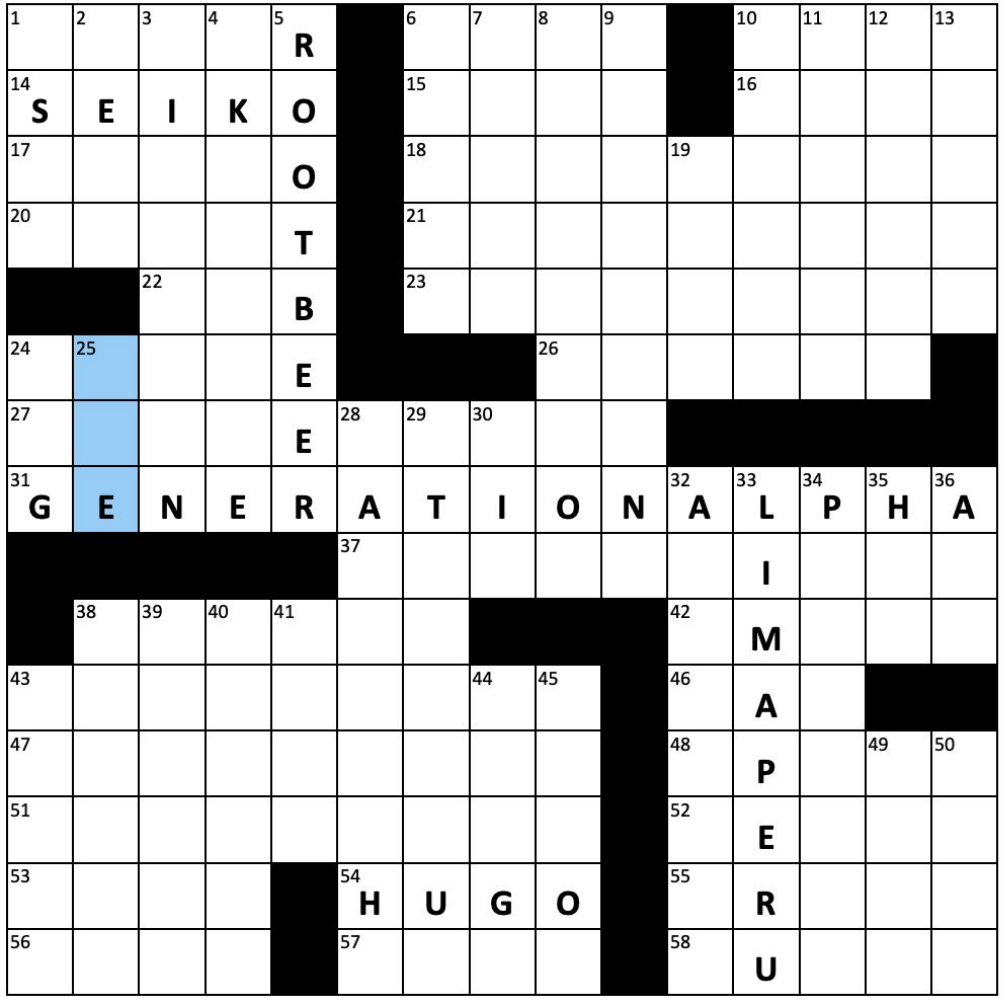
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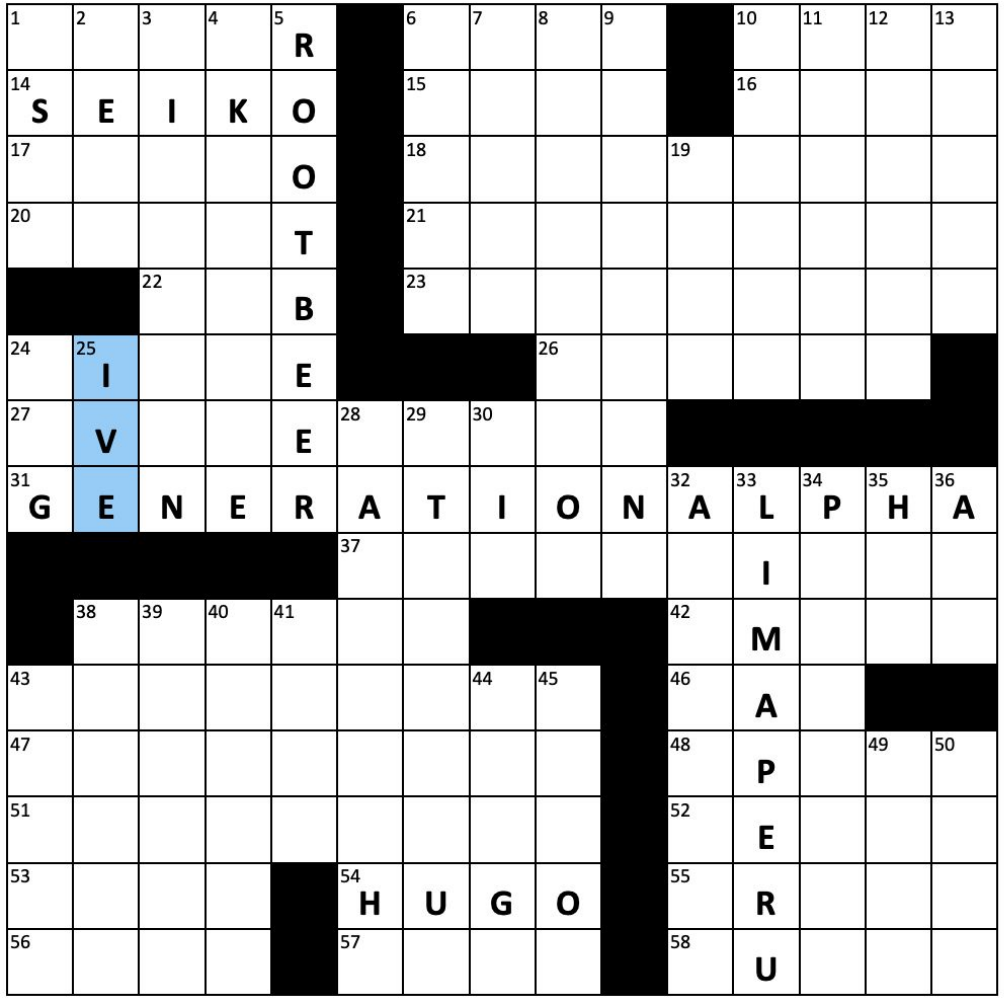
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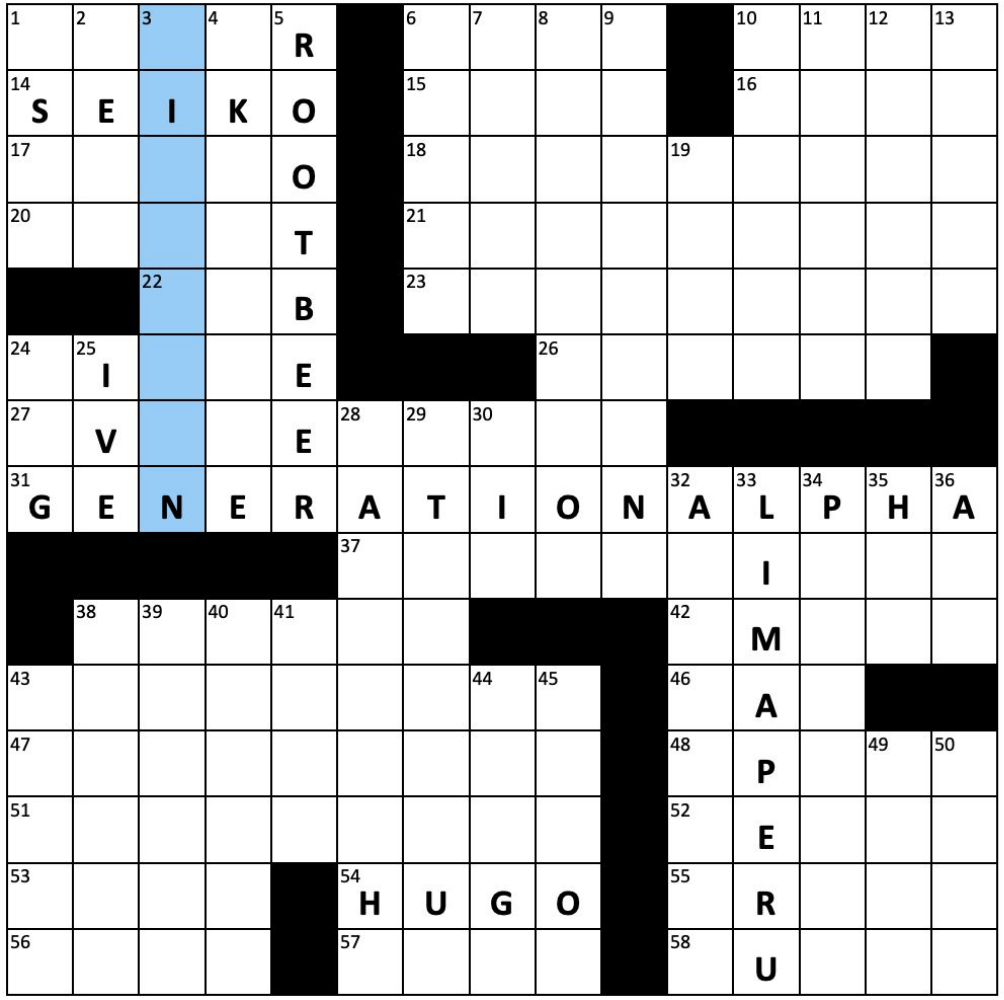
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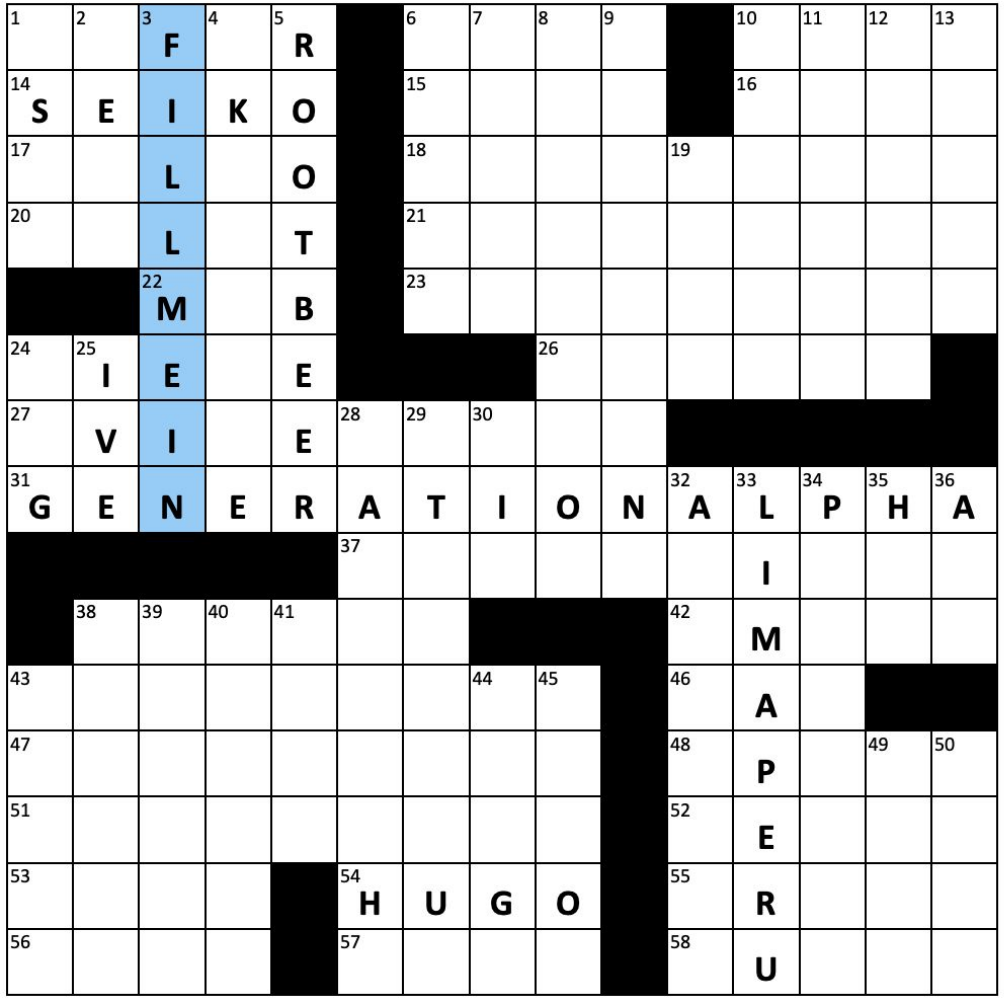
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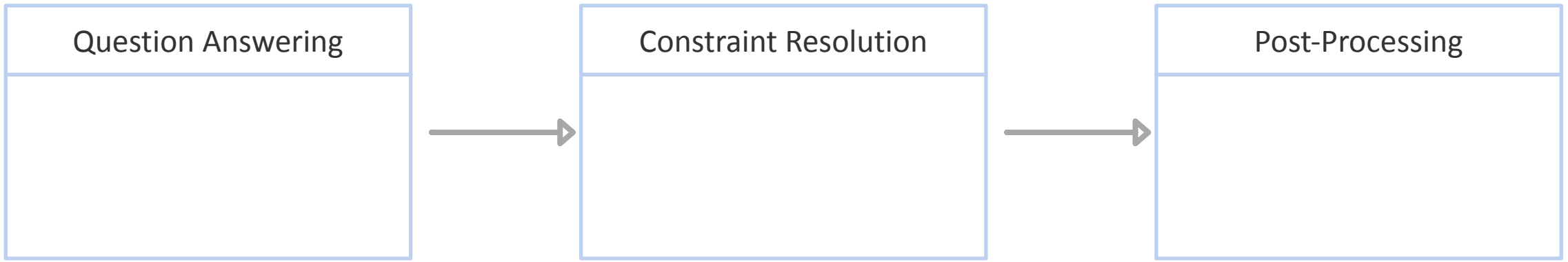
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General Framework





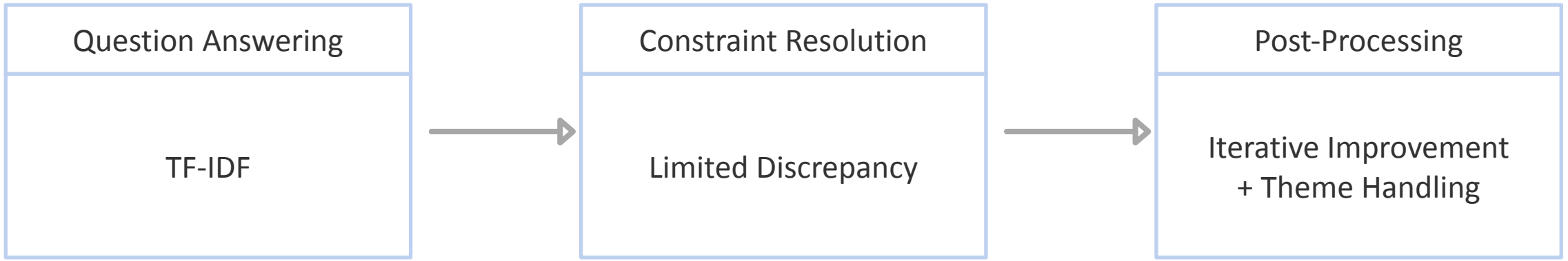
Previous Work

Proverb [Littman et al., 2002]

Question Answering	Constraint Resolution	Post-Processing
ExactMatch, Movies, Writers, Music, Bigram, Encyclopedia, WordNet, …	Belief Propagation + A* Search	

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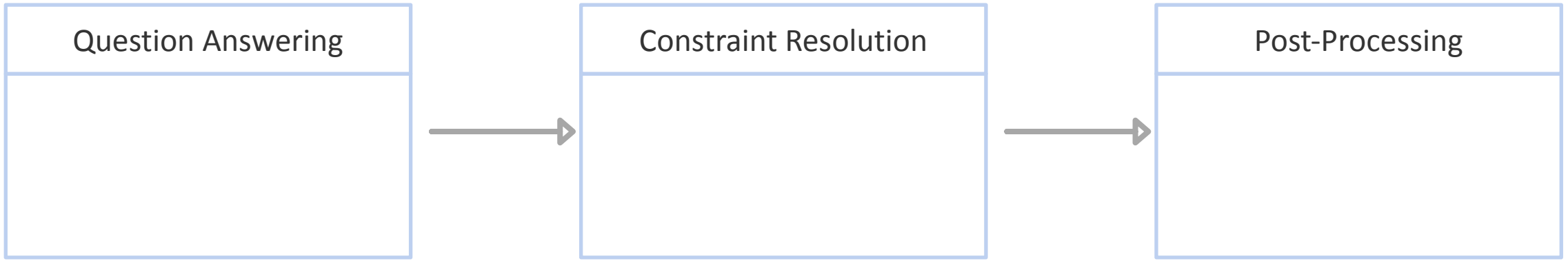
Dr. Fill [Ginsberg, 2011]





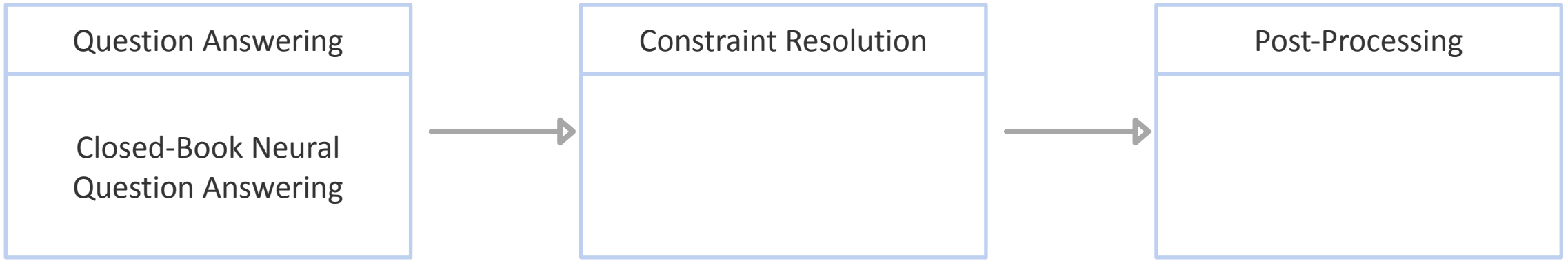
Our Approach

The Berkeley Crossword Solver



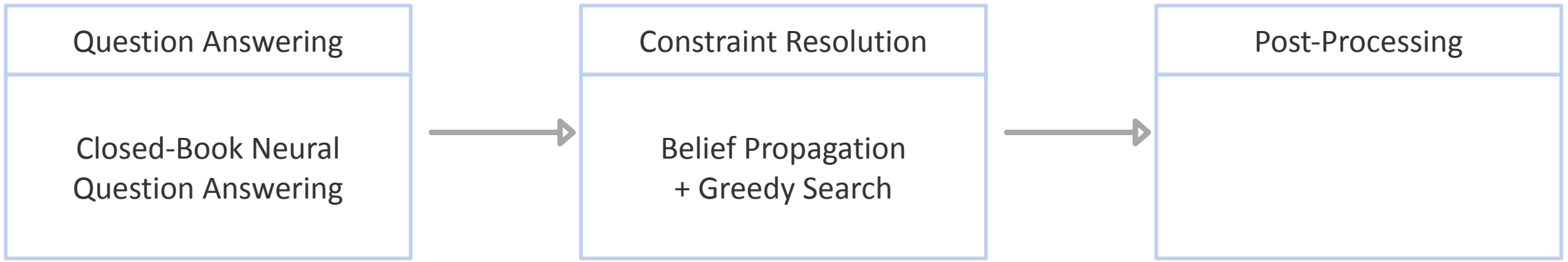
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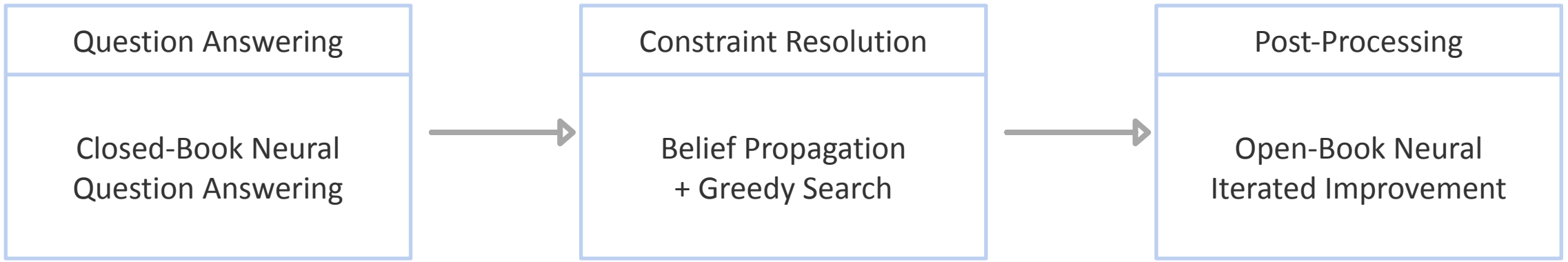
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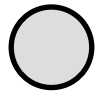
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Question Answering	Constraint Resolution	Post-Processing
Closed-Book Neural Question Answering	Belief Propagation + Greedy Search	Open-Book Neural Iterated Improvement

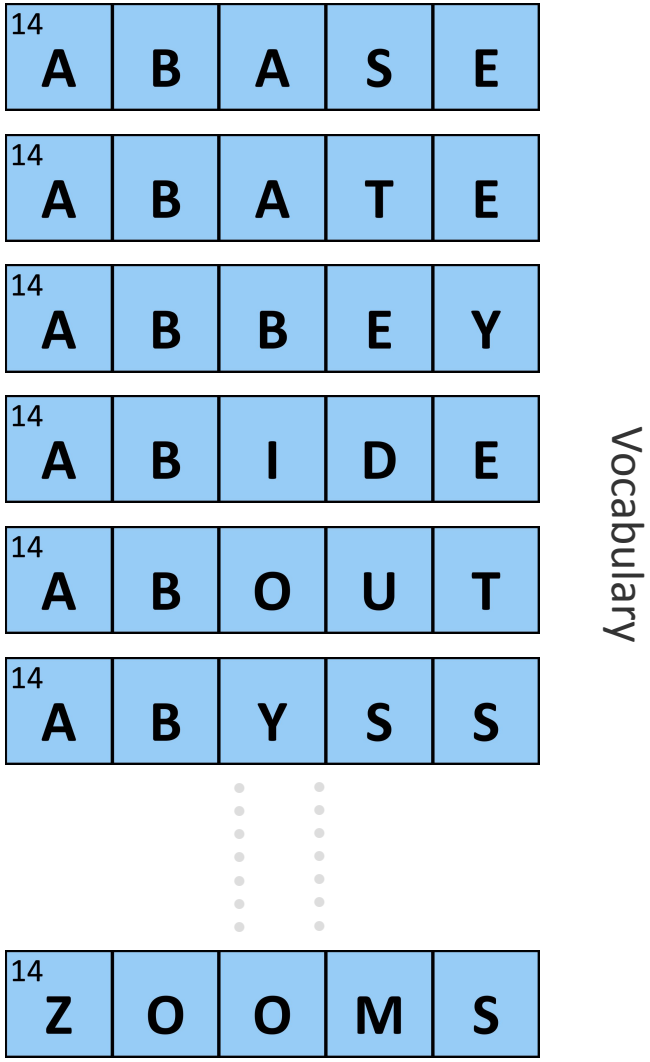
Closed-Book QA

14. Omega competitor



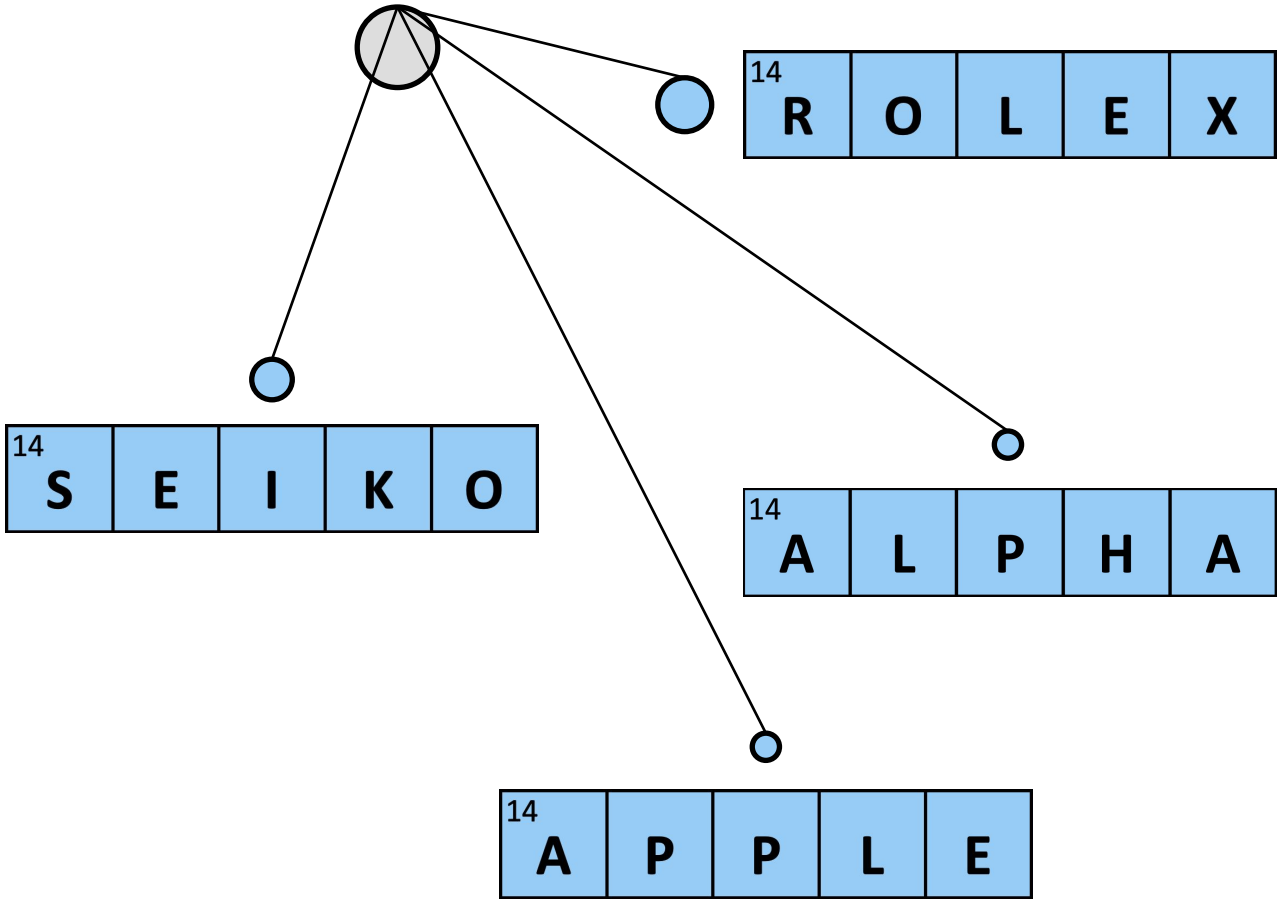
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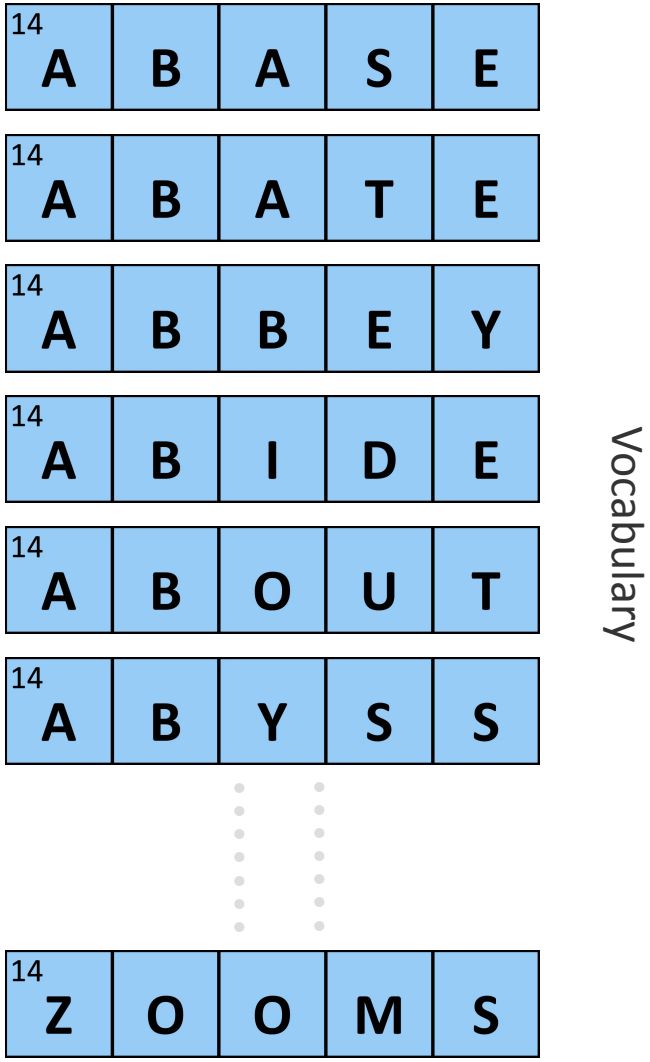
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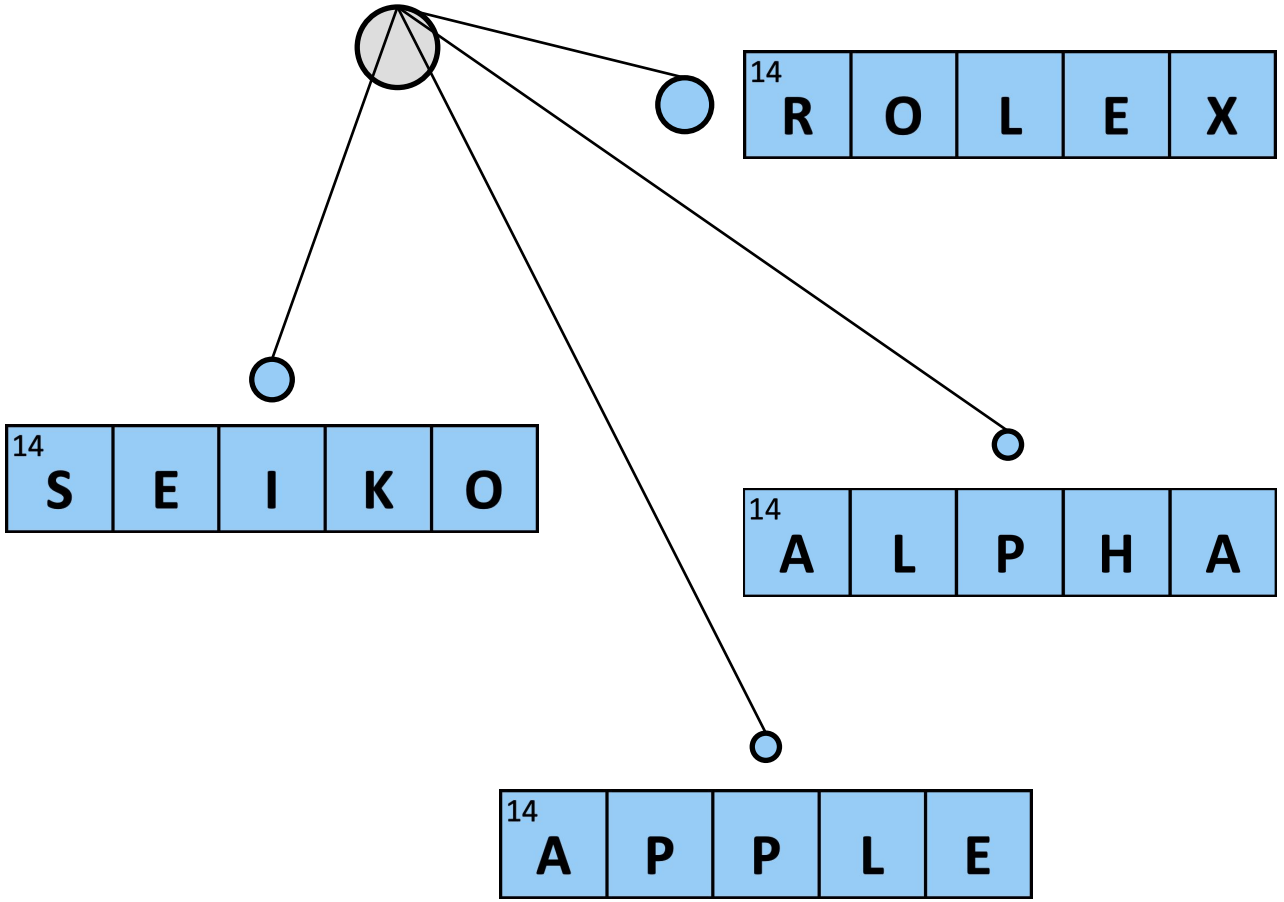
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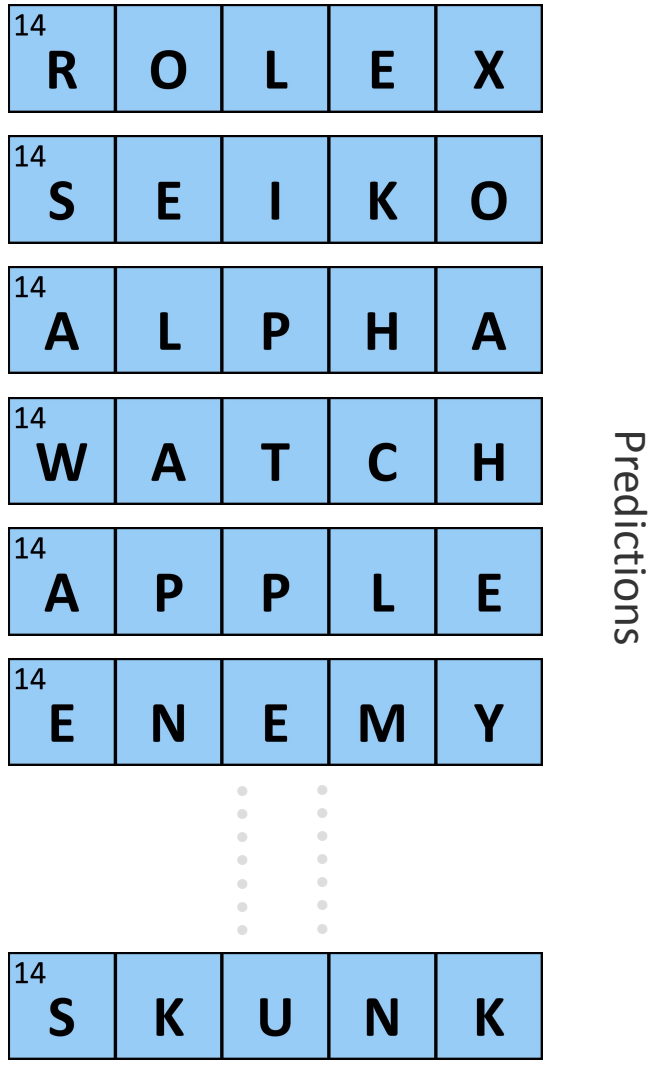




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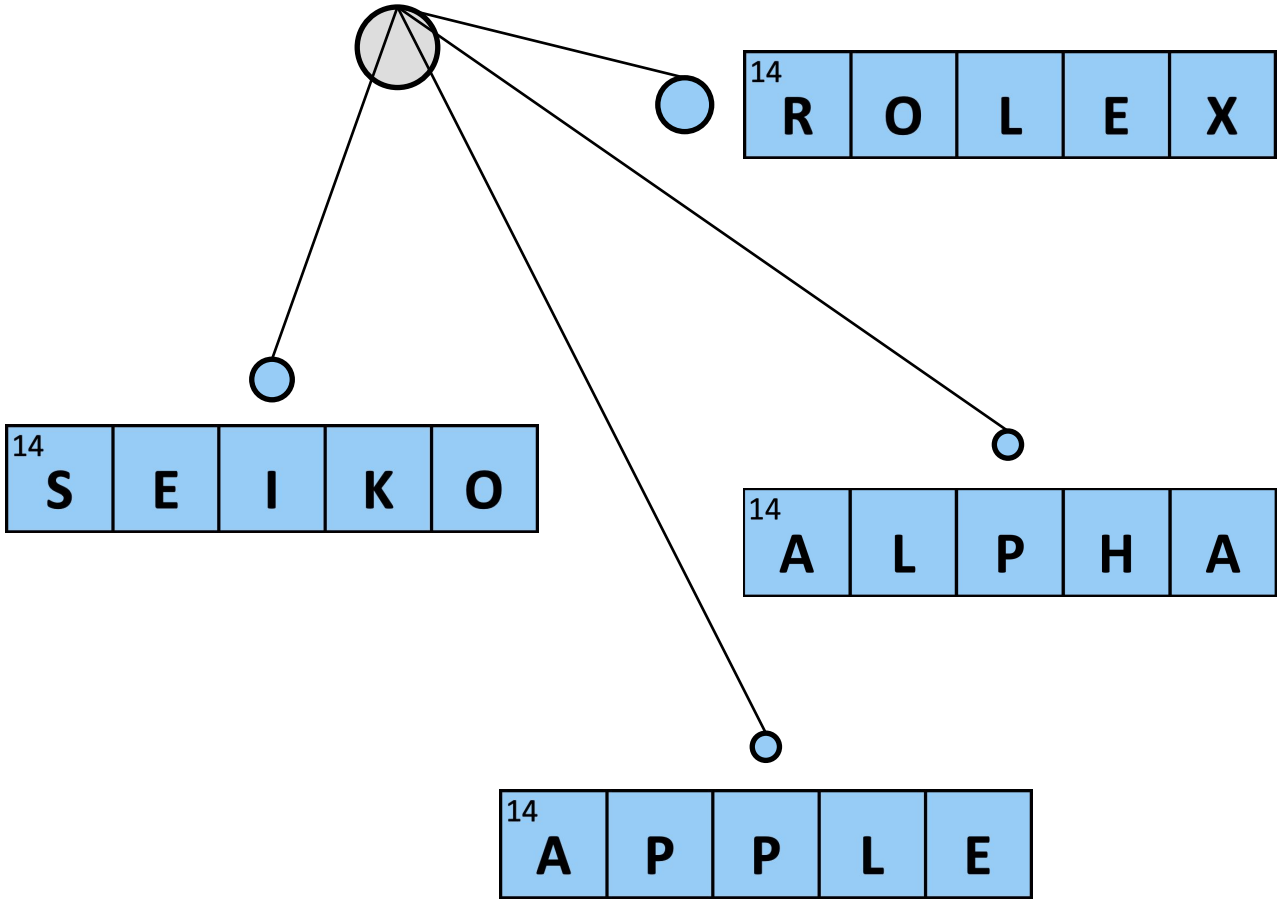
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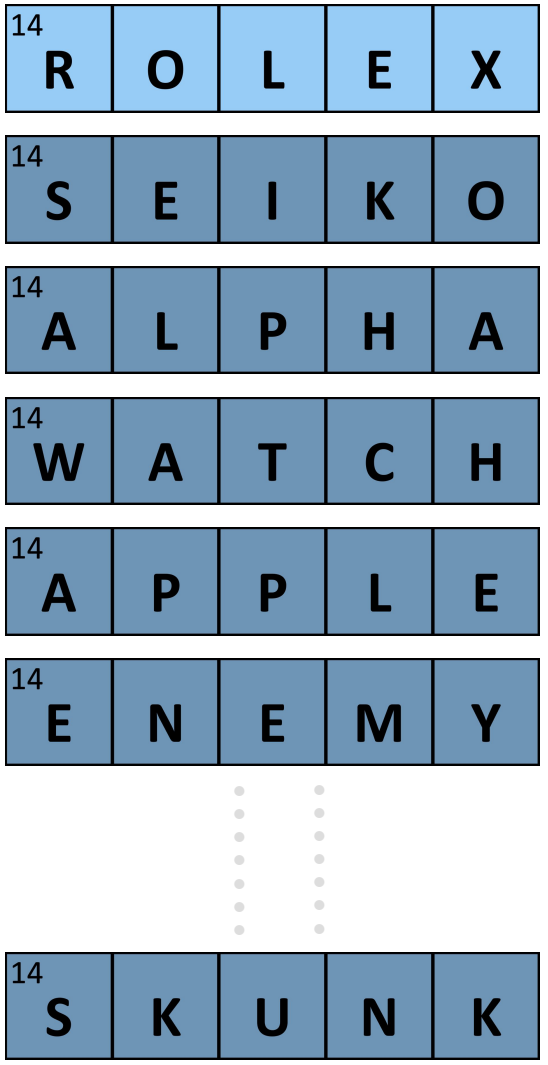




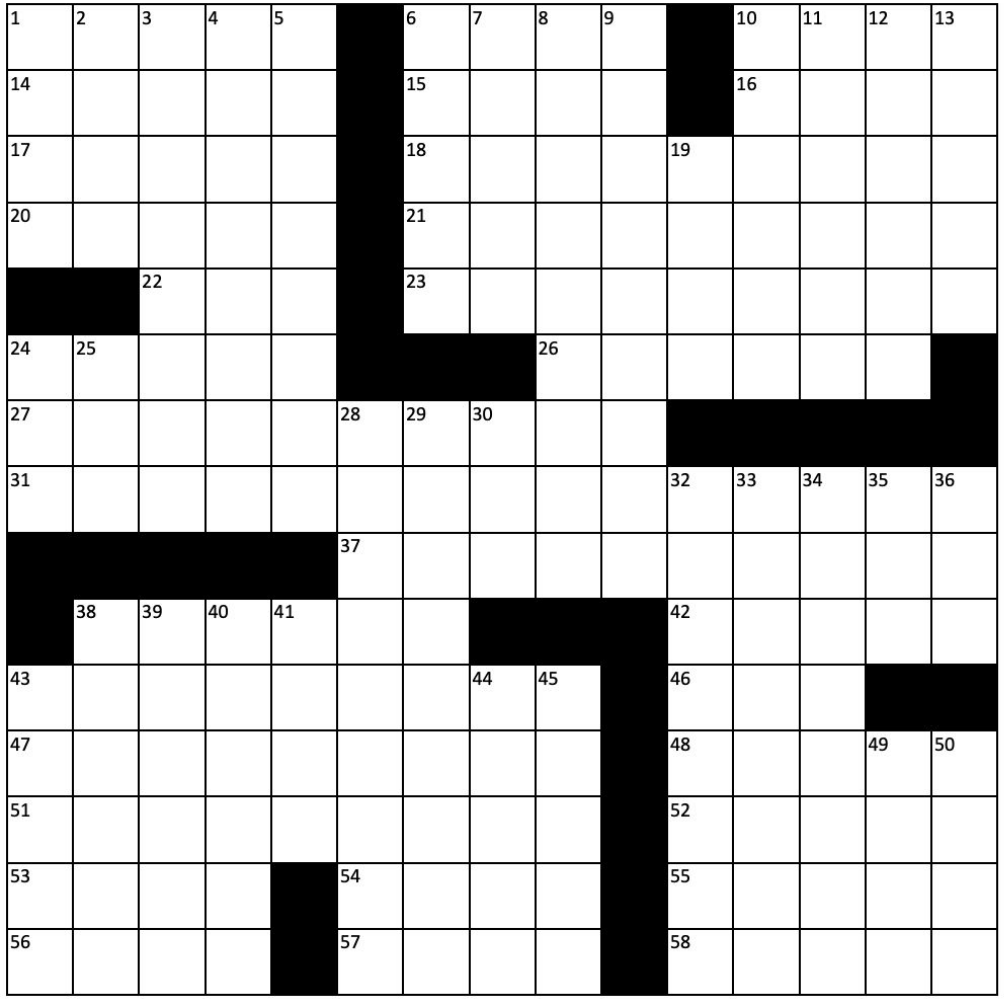
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Handling Ambiguous Clues



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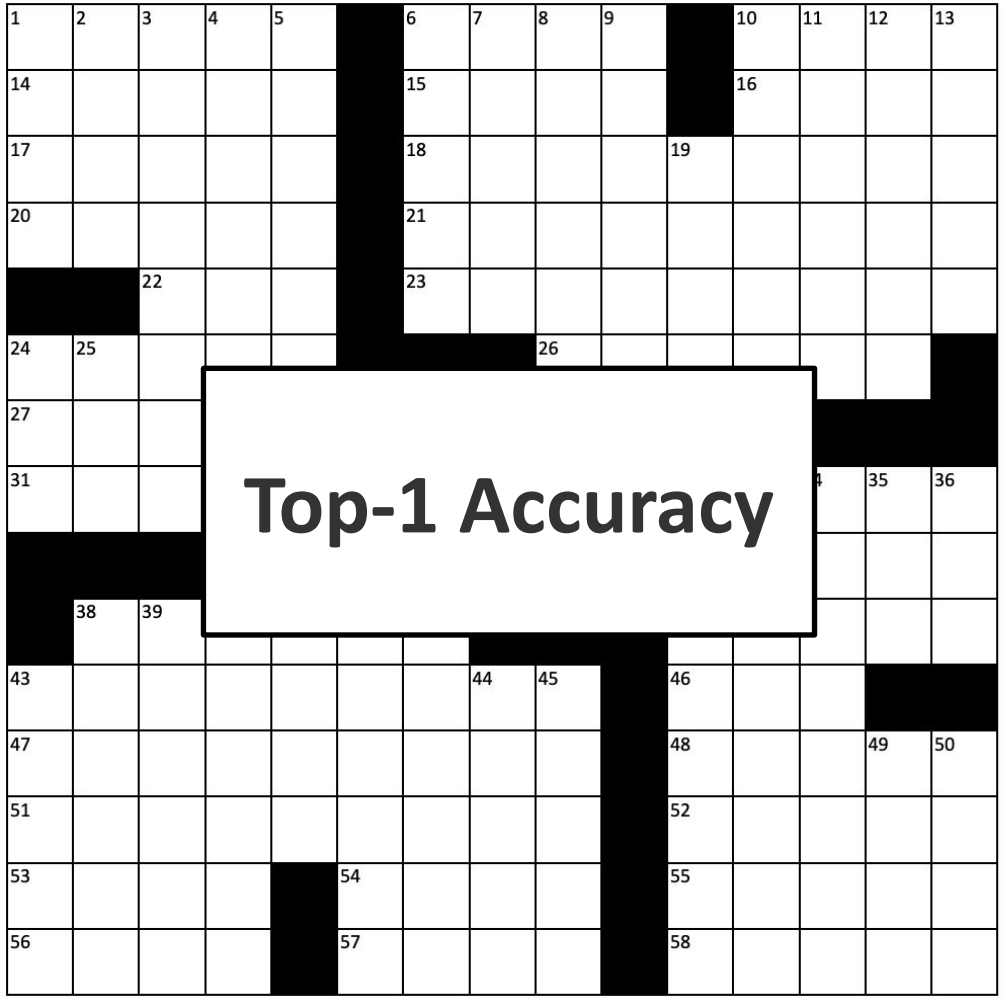
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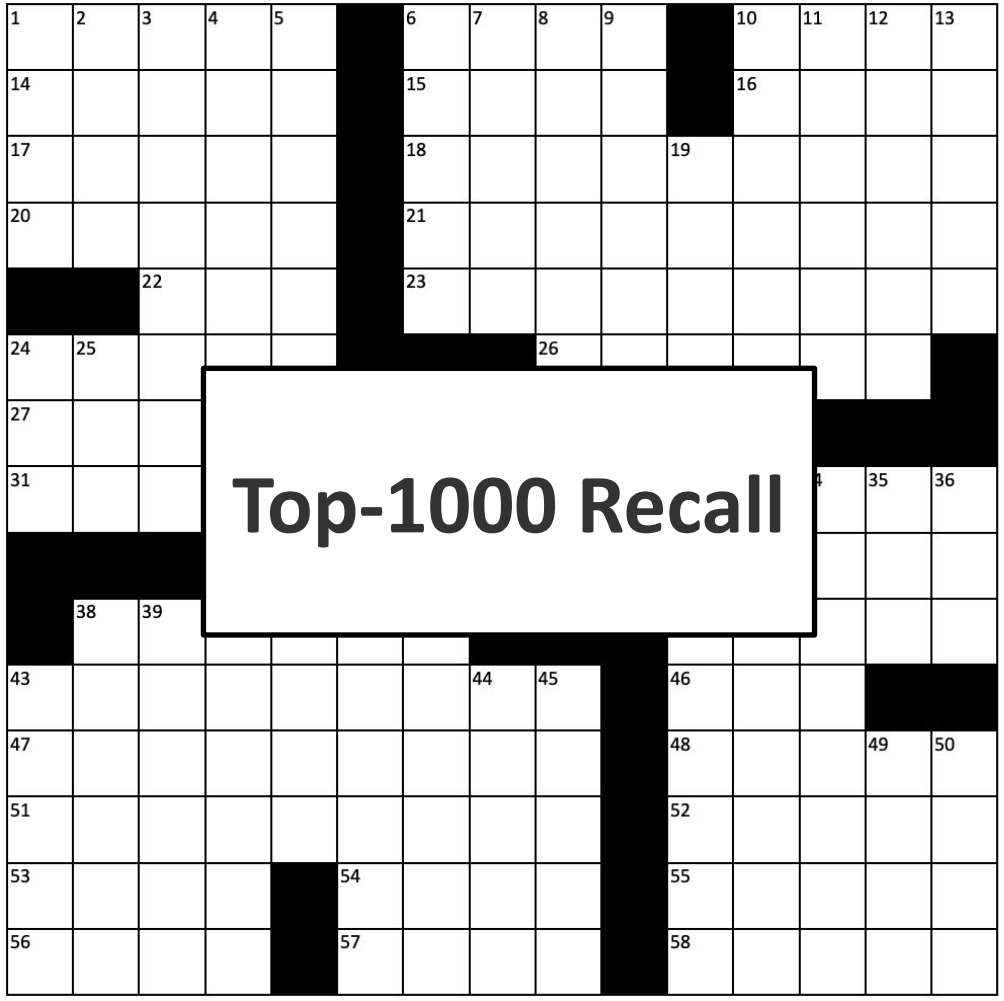
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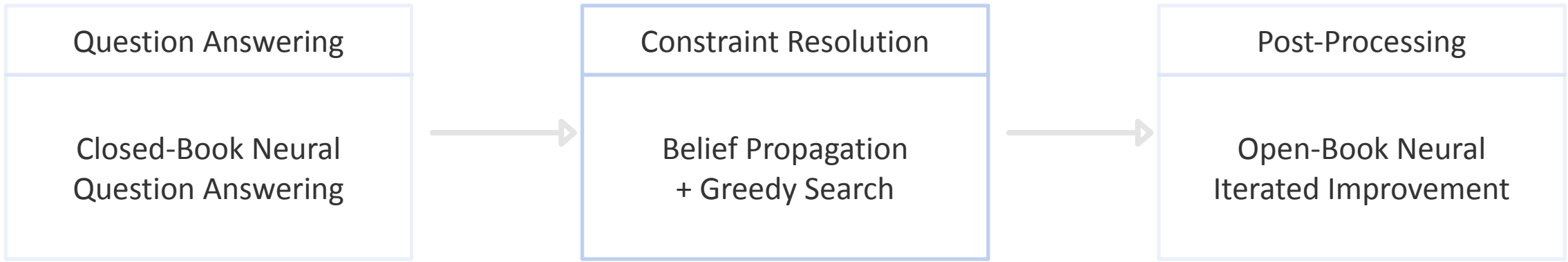
Our Approach

The Berkeley Crossword Solver

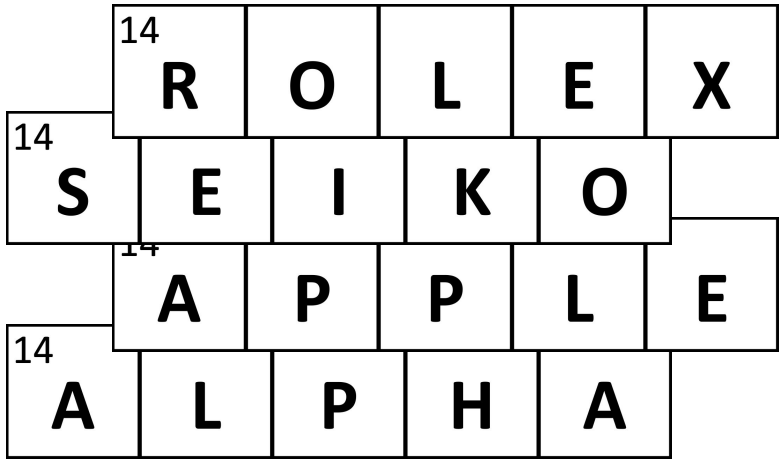
Question Answering	Constraint Resolution	Post-Processing
Closed-Book Neural Question Answering	Belief Propagation + Greedy Search	Open-Book Neural Iterated Improvement

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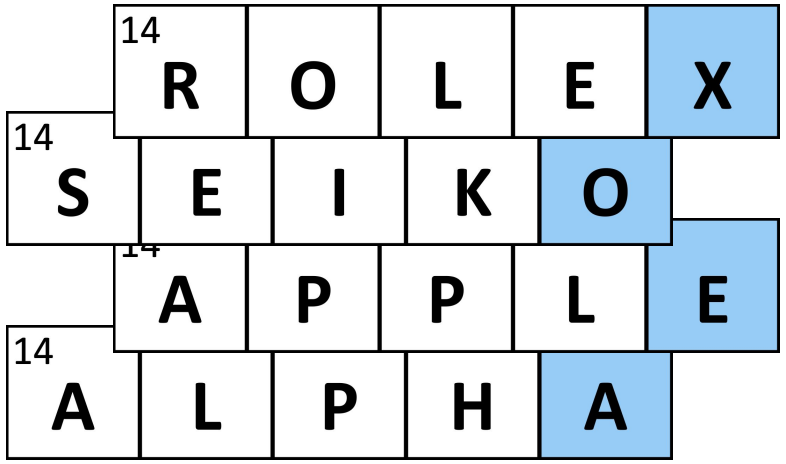
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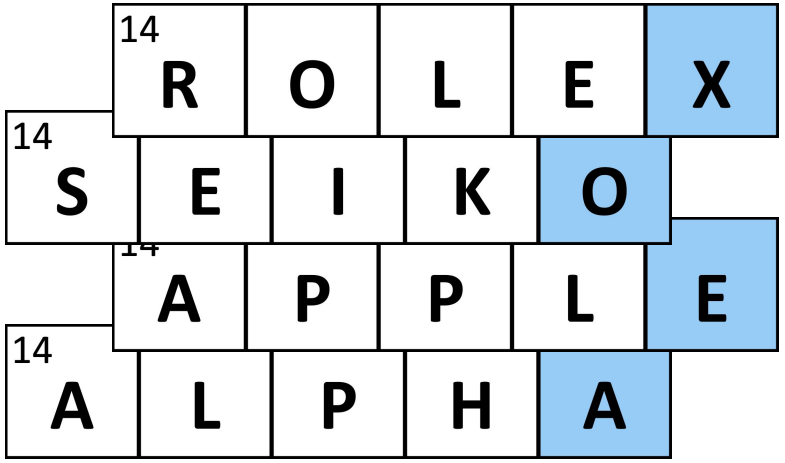
Loopy Belief Propagation

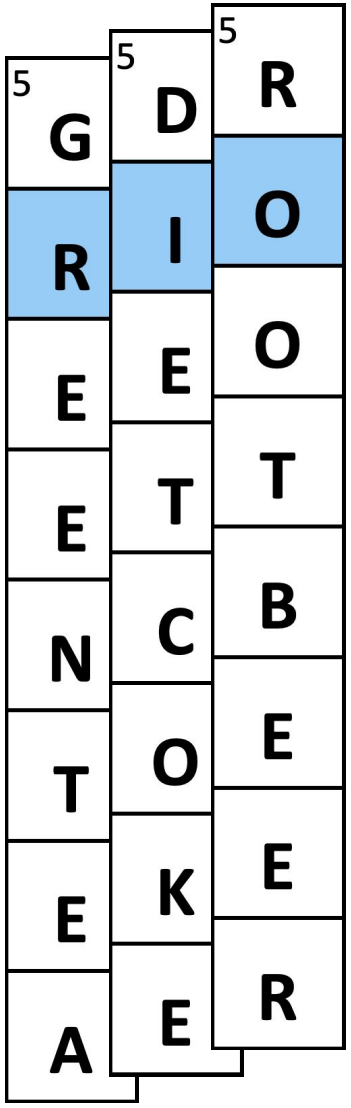


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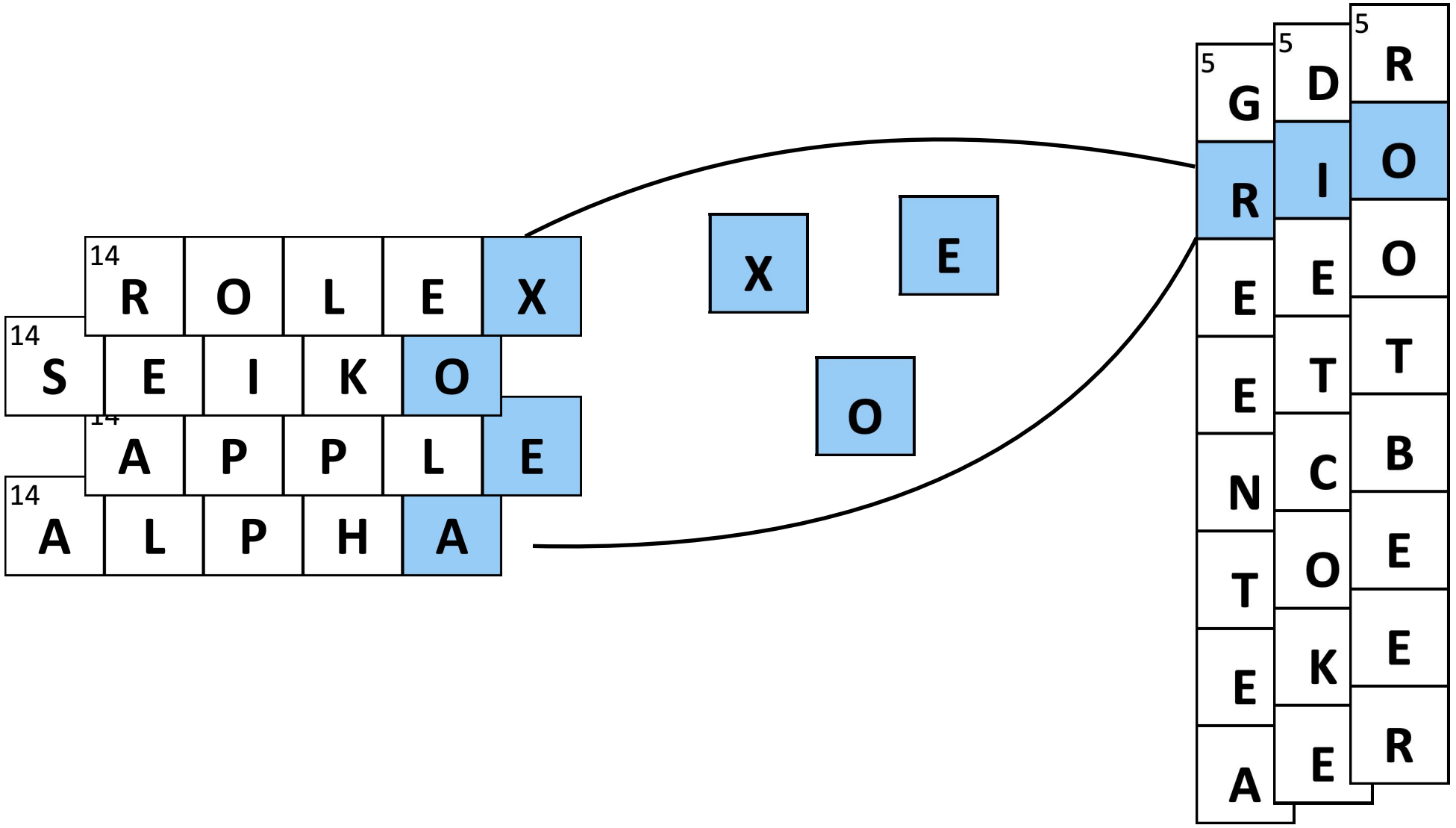


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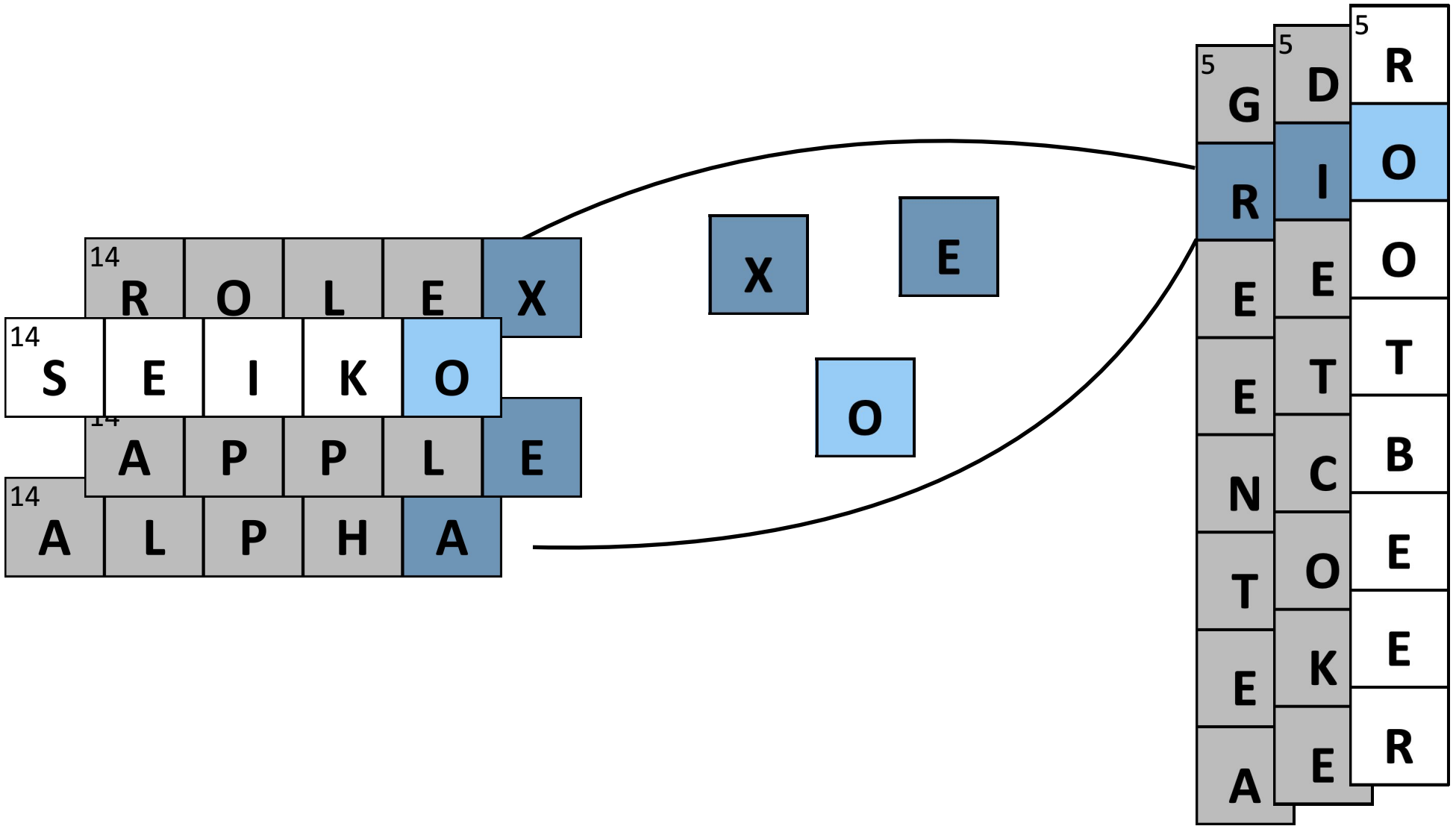




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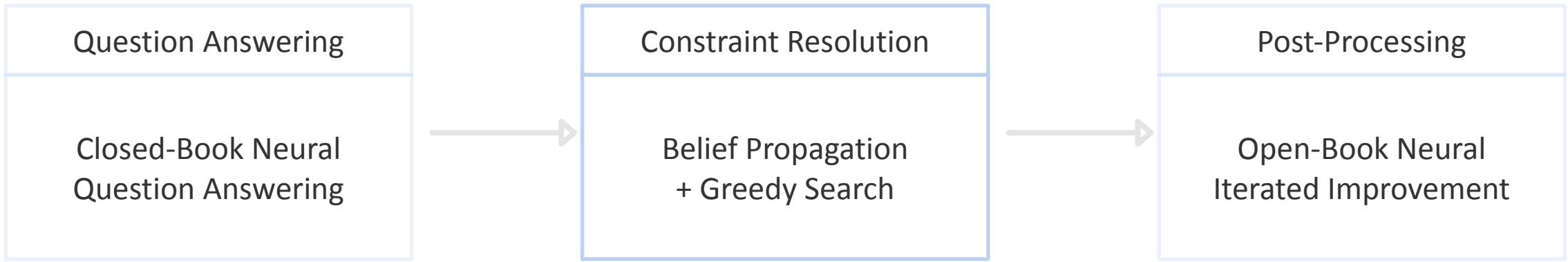


Loopy Belief Propagation



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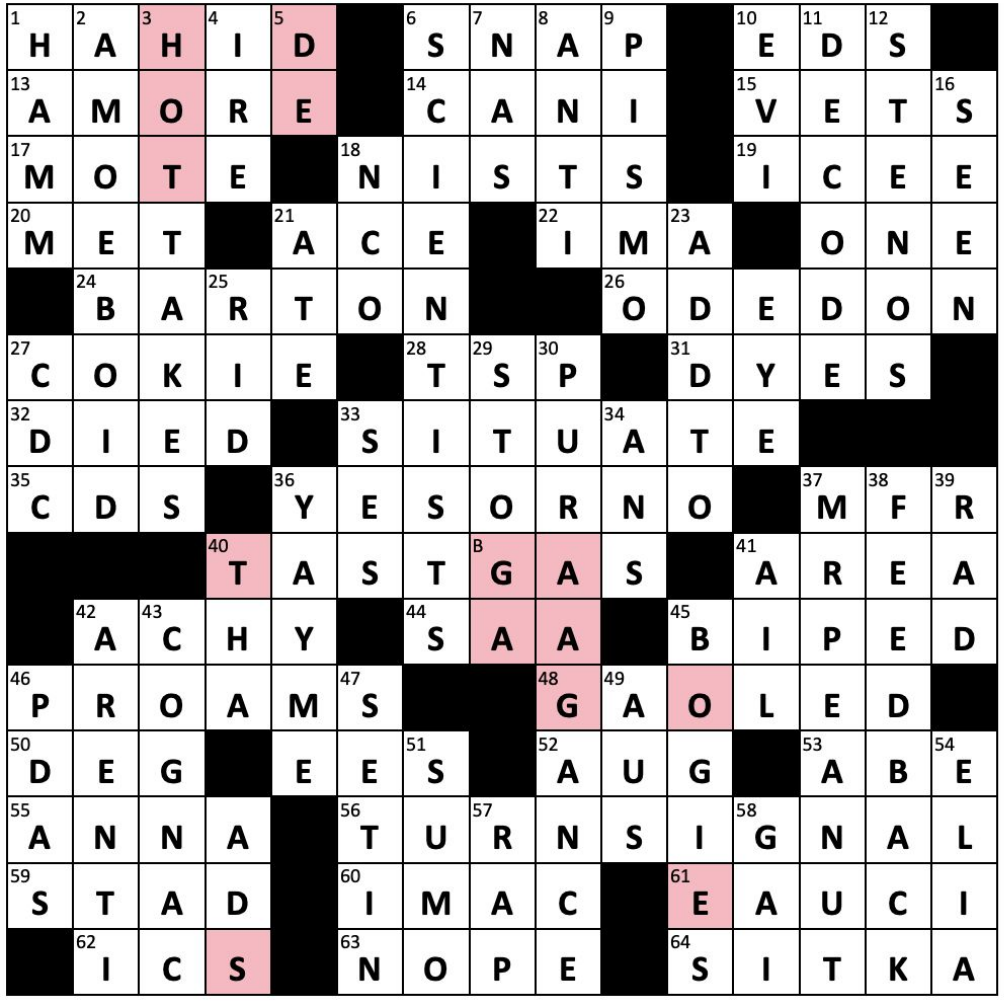


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The Berkeley Crossword Solver

Question Answering	Constraint Resolution	Post-Processing
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Open-Book Improvement



Across

1 Beloved, in Arabic

6 [Make it quick!]

10 Masthead listings, for short

13 Italian herbal liqueur

14 "Please, please, please?"

15 People calling the shots at the zoo?

17 Not radical

19 Drink with a dome-shaped lid

20 Where to find the Egyptian Temple of Dendur, with "the"

21 Crush, as a test

22 Cry from a survivor

24 Clara in the National Women's Hall of Fame

26 Stuffed oneself with, facetiously

27 Newswoman Roberts

28 Recipe amt.

31 Doesn't stay natural?

32 Fell off, as laughter

33 Place

Down

1 Soccer star on a 1999 Wheaties box

2 Bloblike

3 Ill-advised opinions

4 Fury

5 Feeling on a lo-o-ong car trip

6 Lab workers

7 Performer with the hit 2006 album "Hip Hop Is Dead"

8 Voting no

9 ___ Beach, Calif.

10 Satan, with "the"

11 Crack, as a secret message

12 Shorthand writers, for short

16 Dated

18 Cpl. or sgt.

23 Expand

25 Purge (of)

27 Atlanta-based health org.

29 Long fur scarfs

The New York Times Crossword from Thursday, January 21, 2021. Constructed by Daniel Mauer and edited by Will Shortz.

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The 2021 ACPT

American Crossword Puzzle Tournament (ACPT):

- Organized by Will Shortz, NYT Crossword Editor
- Consists of seven crossword puzzles scored based on letter accuracy, word accuracy, puzzle accuracy, and timing

Previous events:

- Proverb: 213th out of 258
- Dr. Fill: 11th out of 618


SLATE

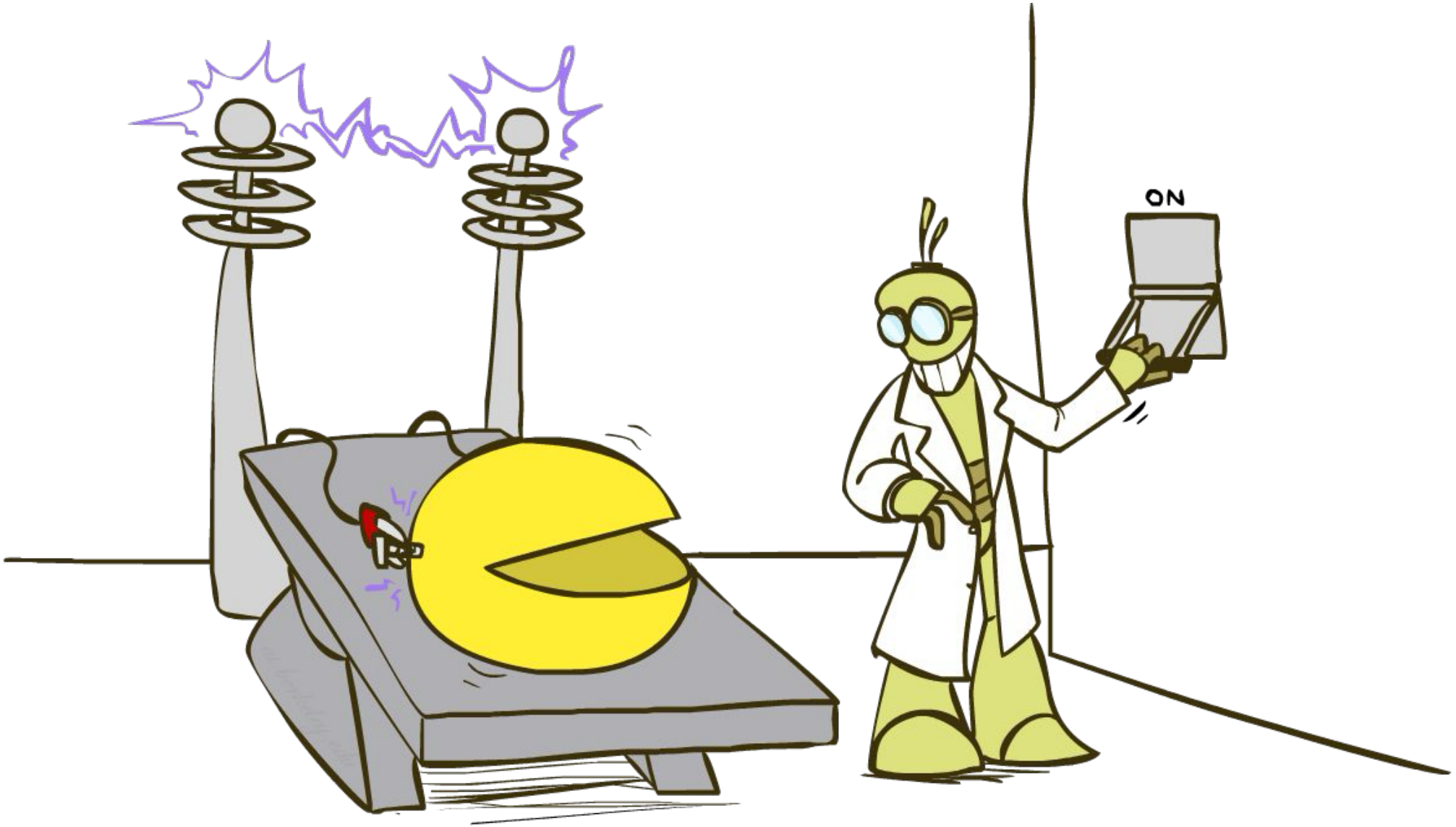
News & Politics Culture Technology Business

SOMETIMES COMMON SENSE IS ENOUGH

USERS

An A.I. Finally Won an Elite Crossword Tournament

Pac-Man Beyond the Game!

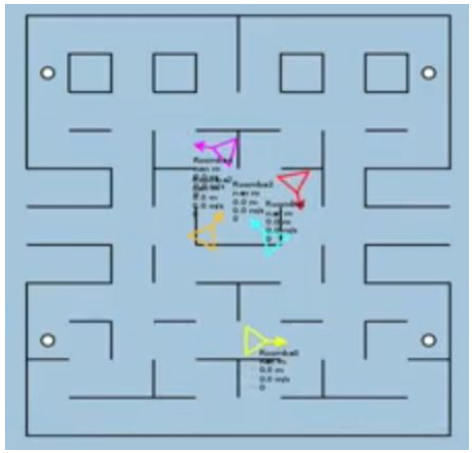


Pacman: Beyond Simulation?





ROOMBA PACMAN





Students at Colorado University: http://pacman.elstonj.com

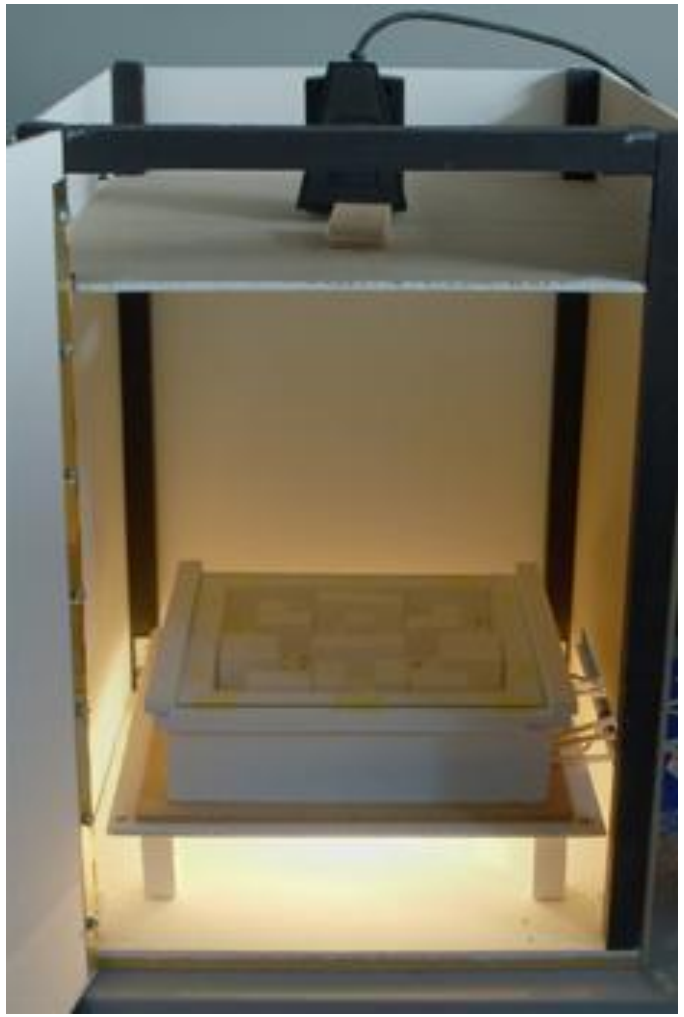
[VIDEO: Roomba Pacman.mp4]

Pacman: Beyond Simulation!



Bugman?

- AI = Animal Intelligence?
  - Wim van Eck at Leiden University
  - Pacman controlled by a human
  - Ghosts controlled by crickets
  - Vibrations drive crickets toward or away from Pacman's location



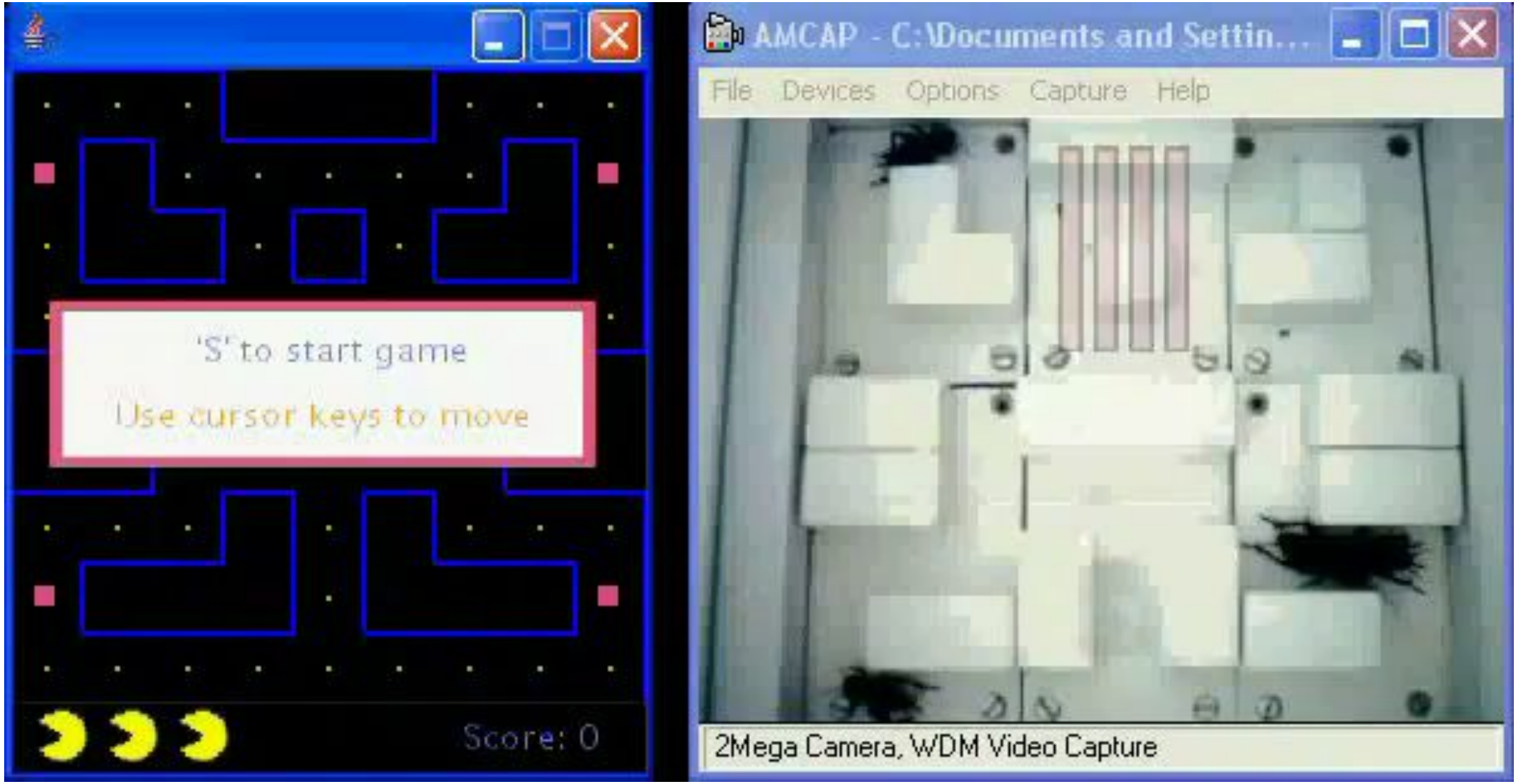




http://pong.hku.nl/~wim/bugman.htm

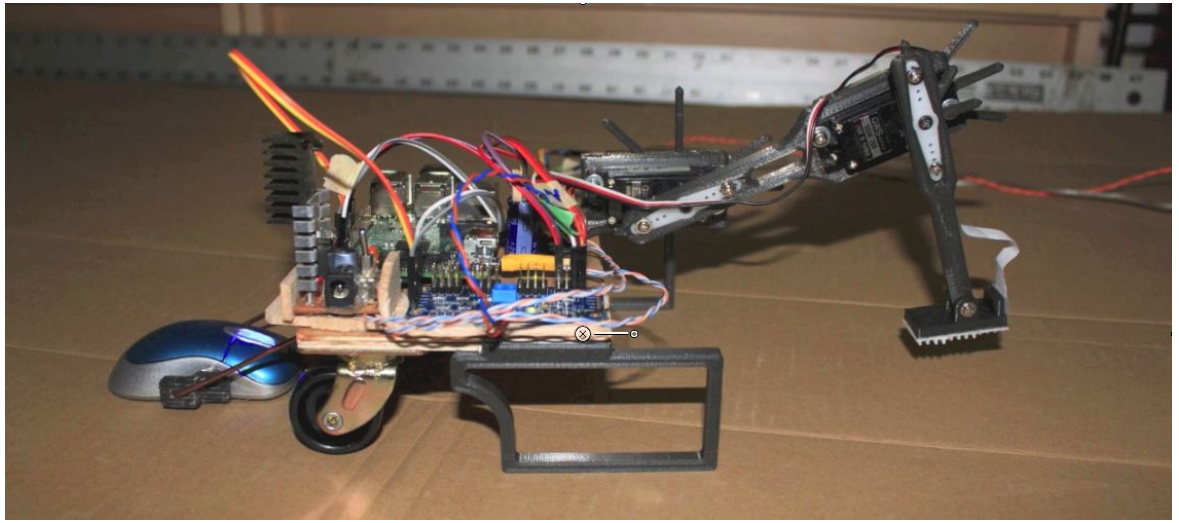
[VIDEO: bugman_movie_1.mov]

Bugman



Crawler





Core AI Ideas

- We've seen many key ideas about how computation can give rise to aspects of intelligent behavior
  - Planning: Search, CSPs, Games (predicting outcomes in simulation)
  - Uncertainty: MDPs, Bayes Nets, Utilities (decision making with uncertainty)
  - Learning: ML and RL: (improving from experience)

- These all touch on different aspects of intelligence
  - The centrality of each has ebbed and flowed
  - But these aspects always seem to return in some form

Core AI Ideas Rising Today

- Reasoning models
- Theory of mind
- Alignment
- …

That’s It!

- Thanks for joining us for this semester!
- Have a great summer and always maximize your expected utilities!